Sanjivani University

SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

UNDERGRADUATE DEGREE PROGRAMME

CURRICULUM AND SYLLABI

Regulation 2024

B. Tech. Computer Science and Engineering (Generic)

Academic Year: 2024-2025

Vision of the Sanjivani University

To emerge as a globally recognized university for excellence in Education, Research, Innovation and Entrepreneurship in order to produce effective leaders for serving the society.

Mission of the Sanjivani University

- Providing state of the art infrastructure, industry centric curriculum and education focusing on ethics and critical thinking.
- Promoting an innovative and research oriented environment for fostering effective problem solving and entrepreneurial development.
- Collaborating with National and International Industries and Higher Education Institutions for academics and research.
- Creating a conducive environment for a vibrant, happy and responsible society by contributing to Institutional Social Responsibility (ISR) initiatives.

Vision of the School of Computer Science and Engineering

To emerge as a globally recognized university for excellence in Education, Research, Innovation and Entrepreneurship in order to produce effective leaders for serving the society.

Mission of the School of Computer Science and Engineering

- Providing state of the art infrastructure, industry-centric curriculum and education focusing on ethics and critical thinking.
- Promoting an innovative and research oriented environment for fostering effective problem solving and entrepreneurial development.
- Collaborating with National and International Industries and Higher Education Institutions for academics and research.
- Creating a conducive environment for a vibrant, happy and responsible society by contributing to Institutional Social Responsibility (ISR) initiatives.

Vision of the Computer Science & Engineering (Generic) Department

To produce top-notch engineers with excellent moral character who can demonstrate leadership by their technical process, creative potential, and useful soft skills, all of which will enhance societal quality of life.

Mission of the Computer Science & Engineering (Generic) Department

- To provide cutting-edge infrastructure and an industry-centric curriculum emphasizing ethics and critical thinking.
- To foster an innovation and research environment, empower students to solve complex problems using “out of the box” thinking.
- To develop full-stack developers with strong technical foundations, analytical skills, and entrepreneurial skills.

Program Outcomes (POs):

1. **Engineering knowledge:** Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
2. **Problem analysis:** Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
3. **Design/development of solutions:** Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, society, and environmental considerations.
4. **Conduct investigations of complex problems:** Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
5. **Modern tool usage:** Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modelling to complex engineering activities with an understanding of the limitations.
6. **The engineer and society:** Apply reasoning in formed by the contextual knowledge to assess social, health, safety, legal and cultural issues, and the consequent responsibilities relevant to the professional engineering practice
7. **Environment and sustainability:** Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
8. **Ethics:** Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
9. **Individual and team work:** Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
10. **Communication:** Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
11. **Project management and finance:** Demonstrate knowledge and understanding of the engineering and management principles and apply the set of one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
12. **Life-long learning:** Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

Program Educational Objectives (PEOs)

1. Graduate will acquire proficiency in developing and executing effective solutions through the utilization of new technologies, platforms, and Free and Open-Source Software (FOSS).

2. Graduate will develop a deep comprehension of the hardware and software tools that meet the current demands of the IT industry.

3. Graduate will understand the contemporary research issues in different areas of computer science & engineering and explore research gaps, analyze and carry out research in the specialized/emerging areas.

Program Specific Objectives (PSOs)

1. Graduate will possess a deep understanding of theoretical computing concepts and practical applications across various domains, including software technology, system programming, data analytics/machine learning, and networking.

2. Graduate will demonstrate the ability to analyze, design, and develop computing systems for diverse engineering applications while upholding professional ethics and social responsibility.

3. Graduate will be proficient in applying computing solutions to address challenges in emerging fields such as Big Data, High-Performance Computing, and Software-Defined Networking, contributing to technological innovation and advancement.

4. Graduate will showcase proficient engineering knowledge and advanced computing skills to tackle complex challenges, devise innovative solutions, and drive technological advancements, thereby contributing significantly to the field of engineering and societal advancement.

COURSE STRUCTURE

B.Tech Computer Science and Engineering

Category	I	II	III	IV	V	VI	VII	VIII	SU	MS-NEP	AICTE
BSC	4	10							14	14-18	23
ESC	8	4							12	12-16	29
AEC	2		2						4	4	-
VSEC	2	2		4					8	8	-
CC	4								4	4	-
MC									0		-
PCC		6	11	11	9	8	4	4	53	44	59
IKS	2								2	2	-
OEC			3	3	3				9	8	9
MDM			2	2	4	3	3		14	14	-
HSSM			0	0					0	0	16
VEC			2	2					4	4	-
PR			2			2			4	2	-
PEC					6	9	9		24	20	12
PW							6		6	4	15
ELC								12	12	14	
Total	22	22	22	22	22	22	22	16	170		163

B.Tech Computer Science and Engineering (Hons.)

Category	I	II	III	IV	V	VI	VII	VIII	SU	MS-NEP	AICTE
BSC	4	10							14	14-18	23
ESC	8	4							12	12-16	29
AEC	2		2						4	4	-
VSEC	2	2		4					8	8	-
CC	4								4	4	-
MC									0		-
PCC		6	11	11	9	8	4	4	53	44	59
IKS	2								2	2	-
OEC			3	3	3				9	8	9
MDM			2	2	4	3	3		14	14	-
HSSM			0	0					0	0	16
VEC			2	2					4	4	-
PR			2			2			4	2	-
PEC					6	9	9		24	20	12
PW							6		6	4	15
ELC								12	12	14	
Total	22	22	22	22	22	22	22	16	170		163
Honours			3	3	3	3	3	3	188	18 +	

B.Tech (Hons). Computer Science and Engineering (Generic) with Double Minor

Category	I	II	III	IV	V	VI	VII	VIII	SU	MS-NEP	AICTE
BSC	4	10							14	14-18	23
ESC	8	4							12	12-16	29
AEC	2		2						4	4	-
VSEC	2	2		4					8	8	-
CC	4								4	4	-
MC									0		-
PCC		6	11	11	9	8	4	4	53	44	59
IKS	2								2	2	-
OEC			3	3	3				9	8	9
MDM			2	2	4	3	3		14	14	-
HSSM			0	0					0	0	16
VEC			2	2					4	4	-
PR			2			2			4	2	-
PEC					6	9	9		24	20	12
PW							6		6	4	15
ELC								12	12	14	
Total	22	22	22	22	22	22	22	16	170		163
Honours			3	3	3	3	3	3	188	18 +	

B.Tech (Hons).Computer Science and Engineering (Generic) with Research

Category	I	II	III	IV	V	VI	VII	VIII	SU	MS-NEP	AICTE
BSC	4	10							14	14-18	23
ESC	8	4							12	12-16	29
AEC	2		2						4	4	-
VSEC	2	2		4					8	8	-
CC	4								4	4	-
MC									0		-
PCC		6	11	11	9	8	4	4	53	44	59
IKS	2								2	2	-
OEC			3	3	3				9	8	9
MDM			2	2	4	3	3		14	14	-
HSSM			0	0					0	0	16
VEC			2	2					4	4	-
PR			2			2			4	2	-
PEC					6	9	9		24	20	12
PW							6		6	4	15
ELC								12	12	14	
Total	22	22	22	22	22	22	22	16	170		163
Research							9	9	188		

B.Tech Computer Science and Engineering (Generic)

SEMESTER-I

Course Code	Course	L/T/P	Contact hrs/week	Credits	Category
24UCSBS101	Mathematics– I	3/1/0	4	4	BSC
24UETIK101	Indian Knowledge System	2/0/0	2	2	IKS
24UETES101	Application Development Practices	3/0/2	5	4	ESC
24UCSES102	Programming and Problem Solving using C++	3/0/2	5	4	ESC
24UETAE101	English- Oral and Written Communication Skills	1/0/2	3	2	AEC
24UETVS101	Design Thinking and Idea Lab	0/0/4	4	2	VSEC
24UETCC101	NSS/YOGA/SPORTS	0/0/8	8	4	CC
MANDATORY COURSE					
24UETMC101	Mandatory Course I(Induction Programme)	Three weeks		1EC	MC
Total			31	22	

SEMESTER II

Course Code	Course	L/T/P	Contact hrs/week	Credits	Category
24UCSBS102	Mathematics – II	3/1/0	4	4	BSC
24UETBS103	Applied Science	4/0/0	4	4	BSC
24UETBS104	Applied Science Laboratory	0/0/4	4	2	BSC
24UCSVS102	Android Programming	0/0/4	4	2	VSEC
24UCSES103	Operating System	3/1/0	4	4	ESC
24UCSPC101	Digital Logic Design & Computer Architecture	2/0/0	2	2	PCC
24UCSPC102	Python Programming	3/0/2	5	4	PCC
MANDATORY COURSE					
24UCSMC102	Mandatory Course II (Sustainable Development)	0/0/2	2	1EC	MC
Total			27	22	

SEMESTER III

Course Code	Course	L/T/P	Contact hrs/week	Credits	Category
24UCSOEXXX	Open Elective 1	3/0/0	3	3	OEC
24UCSMD201	Financial Management	2/0/0	2	2	MDM
24UCSAE201	Analytical Skill Development	2/0/0	2	2	AEC
24UCSVE201	Foreign Language - Japanese/German- I	0/0/4	4	2	VEC
24UCSPR201	Seminar-1	0/0/4	4	2	PR
24UCSPC201	Computer Networks	3/0/0	3	3	PCC
24UCSPC202	Data Structures and Algorithms	3/0/2	5	4	PCC
24UCSPC203	Core Java Programming	3/0/2	5	4	PCC
MANDATORY COURSE					
24UCSMC201	Mandatory Course: III (Soft Skill)	2/0/0		1EC	MC
Total			28	22	

SEMESTER IV					
Course Code	**Course**	**L/T/P**	**Contact hrs/week**	**Credits**	**Category**
24UCSOEXXX	Open Elective-2	3/0/0	3	3	OEC
24UCSVS201	Compiler Construction Tools	2/0/0	2	2	VSEC
24UCSVE202	Foreign Language - Japanese/German- II	0/0/4	4	2	VEC
24UCSVS202	Community Engagement (Mini Project- I)	0/0/4	4	2	VSEC
24UCSMD202	Business Analytics with SAS	2/0/0	2	2	MDM
24UCSPC204	Design and Analysis of Algorithms	3/0/2	5	4	PCC
24UCSPC205	Database Management Systems	3/0/2	5	4	PCC
24UCSPC206	Theory of Computer Science	3/0/0	3	3	PCC
24UCSMC202	Mandatory Course: IV (General Aptitude)	0/0/2	2	0	MC
Total			30	22	

SEMESTER V					
Course Code	**Course**	**L/T/P**	**Contact hrs/week**	**Credits**	**Category**
24UCSOEXXX	Open Elective-3	3/0/0	3	3	OE
24UCSMD301	IPR and EDP	3/1/0	4	4	MDM
24UCSPEXXX	Professional Elective – 1	3/0/0	3	3	PEC
24UCSPEXXX	Professional Elective-2	3/0/0	3	3	PEC
24UCSPC301	Cryptography and Information Security	2/0/2	4	3	PCC
24UCSPC302	Java Enterprise Edition Framework	2/0/2	4	3	PCC
24UCSPC303	PHP & JavaScript	2/0/2	4	3	PCC
		Total	25	22	

SEMESTER VI					
Course Code	**Course**	**L/T/P**	**Contact hrs/week**	**Credits**	**Category**
24UCSMD302	Corporate Laws	3/0/0	3	3	MDM
24UCSPR301	MINI PROJECT-II	0/0/4	4	2	PR
24UCSPEXXX	Professional Elective-3	3/0/0	3	3	PEC
24UCSPEXXX	Professional Elective-4	3/0/0	3	3	PEC
24UCSPEXXX	Professional Elective-5	3/0/0	3	3	PEC
24UCSPC304	Data Mining & Warehousing	3/0/2	5	4	PCC
24UCSPC305	Block Chain Security	3/0/2	5	4	PCC
Total			26	22	

SEMESTER VII					
Course Code	**Course**	**L/T/P**	**Contact hrs/week**	**Credits**	**Category**
24UCSMD401	Research Methodology	3/0/0	3	3	MDM
24UCSPC402	Object Oriented Analysis and Design	3/0/0	3	3	PEC
24UCSPEXXX	Professional Elective-7	3/0/0	3	3	PEC
24UCSPEXXX	Professional Elective-8	3/0/0	3	3	PEC
24UCSPC401	Web Technology using React	3/0/2	5	4	PCC
24UCSPW401	Internship/Project Idea Phase-1	0/0/12	12	6	PW
Total			29	22	

SEMESTER VIII					
Course Code	**Course**	**L/T/P**	**Contact hrs/week**	**Credits**	**Category**
24UCSPC403	Microservices & Distributed Computer Architecture	3/0/2	5	4	PCC
24UCSEL401	Project Work	0/0/24	24	12	ELC
Total			29	16	

BASIC SCIENCE COURSES (14 Credits)

S. No	Course Code	Course Title	L/T/P	Contact Hrs/Wk	Credits	Category
1.	24UCSBS101	Mathematics– I	3/1/0	4	4	BSC
2.	24UETBS102	Mathematics – II	3/1/0	4	4	BSC
3.	24UETBS103	Applied Science	4/0/0	4	4	BSC
4.	24UETBS104	Applied Science Laboratory	0/0/4	4	2	BSC

ENGINEERING SCIENCE COURSES (12 Credits)

S. No	Course Code	Course Title	L/T/P	Contact Hrs/Wk	Credits	Category
1.	24UETES101	Application Development Practices	3/0/2	5	4	ESC
2.	24UCSES102	Programming and Problem Solving usingC++	3/0/2	5	4	ESC
3.	24UCSES103	Operating System	3/1/0	4	4	ESC

AEC (4 Credits)

S. No	Course Code	Course Title	L/T/P	Contact Hrs/Wk	Credits	Category
1.	24UETAE101	English- Oral and Written Communication Skills	1/0/2	2	2	AEC
2.	24UCSAE201	Indian Language (Marathi and Technology)	2/0/0	2	2	AEC

VSEC (8 Credits)

S. No	Course Code	Course Title	L/T/P	Contact Hrs/Wk	Credits	Category
1.	24UETVS101	Design Thinking and Idea Lab	0/0/4	4	2	VSEC
2.	24UCSVS102	Android Programming	0/0/4	4	2	VSEC
3.	24UCSVS201	Software Validation Testing Tools	1/0/2	3	2	VSEC
4.	24UCSVS202	Community Engagement (Mini Project- I)	0/0/4	4	2	VSEC

CC (4 Credits)

S. No	Course Code	Course Title	L/T/P	Contact Hrs/Wk	Credits	Category
1.	24UETCC101	NSS/YOGA/SPORTS	0/0/8	8	4	CC

MANDATORY COURSES

S. No	Course Code	Course Title	L/T/P	Contact Hrs/Wk	Credits	Category
1.	24UETMC101	Mandatory Course I(Induction Programme)	Three weeks		1EC	MC
2.	24UCSMC102	Mandatory Course II (Sustainable Development)	0/0/2	2	1EC	MC
3.	24UCSMC201	Mandatory Course: III (Essence of Indian Traditional Knowledge)	2/0/0		1EC	MC

PROFESSIONAL CORE COURSES (53 Credits)

S. No	Course Code	Course Title	L/T/P	Contact Hrs/Wk	Credits	Category
1.	24UCSPC101	Digital Logic Design	3/0/0	3	3	PCC
2.	24UCSPC102	Python Programming	2/0/2	5	3	PCC
3.	24UCSPC201	Theory of Computer Science	3/0/0	3	3	PCC
4.	24UCSPC202	Data Structures and Algorithms	3/0/2	5	4	PCC
5.	24UCSPC203	Java Programming	3/0/2	5	4	PCC

6.	24UCSPC204	Design and Analysis of Algorithms	3/0/2	5	4	PCC
7.	24UCSPC205	Database Management Systems	3/0/2	5	4	PCC
8.	24UCSPC206	Computer Networks	3/0/0	3	3	PCC
9.	24UCSPC301	Cryptography and Information Security	2/0/2	4	3	PCC
10.	24UCSPC302	Auditing IT infrastructure for Compliance	2/0/2	4	3	PCC
11.	24UCSPC303	Ethical Hacking	2/0/2	4	3	PCC
12.	24UCSPC304	Network and Memory Forensics	3/0/2	5	4	PCC
13.	24UCSPC305	Block Chain Security	3/0/2	5	4	PCC
14.	24UCSPC401	Web Technology using React	3/0/2	5	4	PCC
15.	24UCSPC402	Machine Learning for Cyber Security	3/0/2	5	4	PCC

IKS (2 Credits)

S. No	Course Code	Course Title	L/T/P	Contact Hrs/Wk	Credits	Category
1.	24UETIK101	Indian Knowledge System	2/0/0	2	2	IKS

OEC (9 Credits)

S. No	Course Code	Course Title	L/T/P	Contact Hrs/Wk	Credits	Category
1.	24UCSOEXXX	Open Elective 1	3/0/0	3	3	OEC
2.	24UCSOEXXX	Open Elective-2	3/0/0	3	3	OEC
3.	24UCSOEXXX	Open Elective-3	3/0/0	3	3	OEC

MDM (14 Credits)

S. No	Course Code	Course Title	L/T/P	Contact Hrs/Wk	Credits	Category
1.	24UCSMD201	Financial Management	2/0/0	2	2	MDM
2.	24UCSMD202	Business Analytics with SAS	2/0/0	2	2	MDM
3.	24UCSMD301	IPR and EDP	3/1/0	4	4	MDM
4.	24UCSMD302	Corporate Laws	3/0/0	3	3	MDM
5.	24UCSMD401	Research Methodology	3/0/0	3	3	MDM

VEC (4 Credits)

S. No	Course Code	Course Title	L/T/P	Contact Hrs/Wk	Credits	Category
1.	24UCSVE201	Foreign Language - Japanese/German- I	0/0/4	2	2	VEC
2.	24UCSVE202	Foreign Language - Japanese/German- II	0/0/4	4	2	VEC

PR (4 Credits)

S. No	Course Code	Course Title	L/T/P	Contact Hrs/Wk	Credits	Category
1.	24UCSPR201	Seminar-1	0/0/4	4	2	PR
2.	24UCSPR301	Mini Project-II	0/0/4	4	2	PR

PROFESSIONAL ELECTIVE COURSES (24 Credits)

S. No	Course Code	Course Title	L/T/P	Contact Hrs/Wk	Credits	Category
Software Development						
1.	24UCSPEXXX	Security and Privacy in 6G Networks	3/0/0	3	3	PEC
		Introduction to Automotive Cyber security and Vehicle Networks				
		Cyber security of Industrial Control Systems				
		Cyber Security Threats and Risk: Banking and Finance				
		UI/UX Design				
		Artificial Intelligence				
		Web Frameworks				
		Cloud Computing				
		Frontend Engineering Framework				
		Banking & Insurance				
2.	24UCSPEXXX	Robotics and Cyber Physical Systems	3/0/0	3	3	PEC
		Digital Forensics				
		Deep Fake Prevention				
		Personal Device Forensics				
		Introduction of Telematics for Cyber Security				
3.	24UCSPEXXX	Crypto currency Engineering and Design	3/0/0	3	3	PEC
		Vulnerability Assessment and Penetration Testing				
		Malware Analysis				
		Smart Grid Cyber Security				
		Social Media Overview and Security				
4.		E -Commerce and Digital Payments				
		Cyber security of Drones				

	24UCSPEXXX	Cyber security Management, Compliance and Governance	3/0/0	3	3	PEC
		Security in Wireless Sensor Networks				
		5G Security and Resilience				
5.	24UCSPEXXX	Managing IT Security and Risk	3/0/0	3	3	PEC
		Data Privacy and Security				
		Intrusion Detection Systems				
		Offensive Cyber Security				
		Information Governance and Risk Management				
6.	24UCSPEXXX	Cyberspace Technology and Applications	3/0/0	3	3	PEC
		Applied Cryptography				
		Computer System Security				
		Software Vulnerabilities and Security				
		Cyber Criminology				
7.	24UCSPEXXX	Network Security and Forensics	3/0/0	3	3	PEC
		Incident Response and Audit Compliance				
		Mobile Security and Forensics				
		Intelligent Systems Security				
		IoT Security				
8.	24UCSPEXXX	Drone Forensics	3/0/0	3	3	PEC
		Ethical Hacking				
		Critical Infrastructure Security				
		Web and Database Security				
		Data Analytics for Fraud Detection				
		Cloud Security and Forensics				

PW (6 Credits)

S. No	Course Code	Course Title	L/T/P	Contact Hrs/Wk	Credits	Category
1.	24UCSPW401	Internship/Project Idea Phase-1	0/0/12	12	6	PW

ELC (12 Credits)

S. No	Course Code	Course Title	L/T/P	Contact Hrs/Wk	Credits	Category
1.	24UCSEL401	Project Work	0/0/24	24	12	ELC

SEMESTER - I

24UCSBS101	Engineering Mathematics-I	3/1/0
Nature of Course	Theory	
Prerequisites	Basic Mathematics	
Course Objectives		
1	To make students familiar with concepts and techniques of matrices and their applications such as translation and rotation.	
2	To be able to calculate characteristic polynomial, eigen values and eigen vectors of a given matrix.	
3	To be able to operate within a vector space using the concepts of basis and dimension of a vector space.	
4	To master the techniques of calculus in two and three variables, such as finding critical points.	
5	To understand the analogies between mathematical descriptions of different phenomena in physics and engineering, apply analytical methods and physically interpret the solutions.	
Course Outcomes (Theory) **Upon completion of the course, students shall have ability to**		
1	Determine the solution of a system of linear equations using rank.	[AP]
2	Apply the concept of eigen values and eigen vectors for diagonalizable matrix.	[AP]
3	Calculate basis and dimension of vector space.	[A]
4	Determine extrema of functions with ordinary and partial derivatives.	[AP]
5	Executing the initial value problems for classical PDEs such as heat, wave and Laplace equations.	[AP]

Course Contents

Each unit will contain real time applications using geogebra

UNIT I Matrices **9 Hrs**

Definition and types of matrices, elementary row and column operations, rank of a matrix, system of linear equations and its consistency, solving the system of linear equations.

Case Study: Hands-on

1) Visualization of Elementary Row and Column Operations: Use Geogebra to demonstrate elementary row and column operations like swapping rows, multiplying rows by constants, and adding rows to other rows. Show how these operations affect the matrix visually and how they can be used to simplify matrices.

2) Application in Computer Graphics: Showcase how matrices are used in computer graphics to perform transformations like translation, rotation, scaling, and shearing. Use Geogebra to demonstrate these transformations visually, allowing students to interactively manipulate objects and see how matrices affect their positions and shapes.

UNIT II Eigen Values and Eigen Vectors **9 Hrs**

Eigen Values, Eigen Vectors and their basic properties, Eigen Space, Diagonalization of a matrix, Cayley-Hamilton theorem.

Case Study: Hands-on

1) Visualizing Eigenvalues and Eigenvectors: Using Geogebra applet where students can input a matrix, and the applet visually illustrates the eigenvalues and eigenvectors of the matrix.

2) Application in Image Processing: Showcase how eigenvalues and eigenvectors are used in image processing for tasks like image compression and denoising.

UNIT III Vector Spaces **9 Hrs**

Vector Spaces, Subspaces, linear span, linear dependence and independence of vectors, basis and dimension of a vector space.

Case Study: Hands-on

Visualizing Vector Spaces and Subspaces: Allow students to input vectors and observe how they span a vector space. Show how adding or removing vectors affects the span and demonstrate the concept of subspaces by restricting the span to certain dimensions.

UNIT IV Functions of Several Variables **9 Hrs**

Continuity, partial derivatives, Jacobian, composite functions(chain rule), Euler's theorem on homogeneous function, Mean Value Theorem(without proof), Taylor's theorem, Extrema, critical and saddle points.

Case Study: Hands-on

1) Visualizing Continuity and Partial Derivatives: Allow students to input multivariable functions and observe how changes in one variable affect the function's continuity and partial derivatives.
2) Finding Extrema and Critical/Saddle Points: Allow students to input a function and visualize its contour plot. Identify and mark the extrema and critical/saddle points on the plot, and provide numerical values for these points.

UNIT V Ordinary and Partial Differential Equations **9 Hrs**

Exact Differential Equations, Linear Differential Equations, Variable Separable Method, PDE- Definition and types, Heat, Wave and Laplace Equations.

Case Study: Hands-on

Geogebra demonstrations for each type of PDE – heat, wave, and Laplace equations: Provide examples of physical systems modeled by each equation (e.g., heat conduction, wave propagation, and steady-state temperature distribution) and show how to solve them using appropriate techniques such as separation of variables.

Total Hours:	**45**

Text Books	
1	B. S. Grewal, Higher Engineering Mathematics, 44/e, Khanna Publishers, 2023-24
2	H. K. Das, Advanced Engineering Mathematics, S Chand, 2019.
3	Mathematics I Calculus and Linear Algebra [For Computer Science Engineering Branches] | AICTE Prescribed Textbook (English), 2022
Reference Books	
1	N. P. Bali and Manish Goyal, A Text Book of Engineering, Mathematics, 11/e, Lakshmi Publications, 2018.
2	D. C. Lay, S. R. Lay and J. J. MacDonald, Linear Algebra and its Applications, Pearson, 2016.
3	B. V. Ramana, Higher Engineering Mathematics, Tata McGraw-Hill, 2017.
4	Paul Dawkins, Calculus III, Lamar University Texas, 2018,
Web References	
1.	https://math.mit.edu/classes/18.03/sup/sup0.pdf
2.	http://tutorial.math.lamar.edu
3.	https://bmsce.ac.in/Content/MAT/Typical_Examples_of_UNIT-2.pdf
Online Resources	

1	https://www.youtube.com/watch?v=UlFC7ErUyOQ&list=PLtfK3oO7bEagOrzs7EW7_UjLx05ZS-KhJ&pp=iAQB
2	https://www.youtube.com/watch?v=0WZ0NiqgoHA&list=PLtfK3oO7bEaiBTQoncjzQ5tWmvDVR5yFU&pp=iAQB
3	https://www.yotube.com/watch?v=6RQd-bE0CnU&list=PLUDxRrqxbYolspoW_D2zIghXWtNKCfQyf&pp=iAQB

24UETIK101	Indian Knowledge System	2/0/0
Nature of Course	Theory	
Prerequisites	NIL	
Course Objectives		
1	To learn Indian Knowledge System with Vedic Corpus	
2	To understand philosophical systems and wisdom through the Ages	
3	To understand knowledge framework and classification with Linguistics	
4	To learn Number Systems and Units of Measurement and Health wellness	
5	To understand town planning and architecture, governance and public administration.	
Course Outcomes (Theory) - Upon completion of the course, students shall have ability to		
1	Understand Indian Knowledge System with Vedic Corpus.	[U]
2	Understand philosophical systems and wisdom through the Ages	[U]
3	Understand knowledge framework and classification with Linguistics	[U]
4	Understand Number Systems and Units of Measurement and Health wellness	[U]
5	Understand town planning and architecture, governance and public administration.	[U]

Course Contents

Unit I Indian Knowledge and The Vedic Corpus **6 Hrs**

Introduction of Indian Knowledge System with its organization, history and some salient aspects. Introduction to Vedas, A synopsis of the four Vedas, Sub-classification of Vedas, Messages in Vedas, Introduction to Vedāṅgas, Prologue on Śikṣā and Vyākaraṇa, Basics of Nirukta and Chandas, Introduction to Kalpa and Jyotiṣa Vedic Life,A Distinctive Features

Case Study

Preservation of Vedas through Oral Tradition,Analyze the techniques used to preserve the Vedas and their impact on cultural continuity.

Unit II Philosophical Systems and Wisdom through the Ages **6 Hrs**

An introduction to philosophical systems, Development of philosophy, Unique features of philosophy, Sāṅkhya approach of philosophy, Introduction to Yoga, Tenet of Nyāya philosophy, Principles of Vaiśeṣika, Gateways of ancestral wisdoms, Introduction to Purāṇa, The Purāṇic repository, Issues of interest in Purāṇas, Introduction to Itihāsas, Key messages in Itihāsas, Wisdom through Nīti-śāstras, Wisdom through Subhāṣita

Case Study

Relevance of Bhagavad Gita in Modern Life: Analyze the teachings of the Bhagavad Gita and their application in contemporary personal and professional life.

Unit III Knowledge Framework and classifications and Linguistics – **6 Hrs**

Indian scheme of knowledge, The knowledge triangle, **Prameya** : A vaiśeṣikan approach to physical reality, **Dravyas :** the constituents of the physical reality, **Attributes:** the properties of substances and Action the driver of conjunction and disjunction, Sāmānya, viśēṣa, samavāya, **Pramāṇa :** the means of valid knowledge, **Saṃśaya:** ambiguities in existing knowledge, Framework for establishing valid knowledge, Deductive or inductive logic framework, Potential fallacies in the reasoning process, **Siddhānta:** established tenets in a field of study, Introduction to Linguistics, Aṣṭādhyāyī, Phonetics, Word generation, Computational aspects

Case Study

Computational Linguistics in Ancient India: Examine how ancient Indian linguists approached problems in

computational linguistics and its relevance today.

Unit IV Number Systems and Units of Measurement with Health Wellness and Psychology 6 Hrs

Number systems in India, Historical evidence, Salient aspects of Indian Mathematics, Bhūta-Saṃkhyā system, Kaṭapayādi system, Measurements for time, distance, and weight, Piṅgala and the Binary system, Introduction to health, **Sapta-dhātavaḥ**: seven-tissues, Role of agni in health, Tri-doṣas, **Āyurveda:** definition of health, Indian approach to psychology, The tri guṇa system & holistic picture of the individual, The Nature of Consciousness, Consciousness studies and issues.

Case Study

Application of Āyurvedic Principles in Modern Healthcare: Analyze how Āyurvedic concepts are integrated into modern wellness practices.

Unit IV Town Planning and Architecture with Governance and Public Administration 6 Hrs

Perspective of Arthaśāstra on town planning, **Vāstu-śāstra** The science of architecture, Eight limbs of Vāstu, Town planning, Temples in India: marvelous stone architecture for eternity, Temple architecture in India, Iconography, Introduction to raja dharma, **Arthaśāstra:** a historical perspective, Elements of a kauṭilyan state, The king & the amātya, Janapada & durga 6. Treasury and the State Economy (Kośa), Danda, Mitra, The Administrative Setup, Relevance of Arthaśāstra, Public Administration in Epics.

Case Study

Relevance of Kautilya's Arthaśāstra in Modern Governance: Examine the principles of Arthaśāstra and their application in contemporary administrative practices.

Total Hours:	**30**

Text Books:	
1.	Mahadevan, B., Bhat Vinayak Rajat, Nagendra Pavana R.N. (2022), "Introduction to Indian Knowledge System: Concepts and Applications", PHI Learning Private Ltd. Delhi.
Reference Books:	
1.	Samskrita Bharati,(2020) "Pride of India: A Glimpse into India's Scientific Heritage" , New Delhi.
2.	Sampad and Vijay (2011). "The Wonder that is Sanskrit", Sri Aurobindo Society, Puducherry.
3.	Acarya, P.K. (1996). Indian Architecture, Munshiram Manoharlal Publishers, New Delhi.
4.	Kapoor Kapil, Singh Avadhesh (2021). "Indian Knowledge Systems Vol – I & II", Indian Institute of Advanced Study, Shimla, H.P.
5.	Dasgupta,S. (1975). A History of Indian Philosophy- Volume 1, Motilal Banarsidass, New Delhi.
6.	PLofker, K. (1963). Mathematics in India, Princeton University Press, New Jeresy, USA.
Web References:	
1.	Ministry of Education, Government of India - https://www.education.gov.in/nep/indian-knowledge-systems
Online Resources:	
1.	Swayam – https://onlinecourses.swayam2.ac.in/imb23_mg55/preview

24UETES101	Application Development Practices	3/0/2
Nature of Course	Theory & Programming	
Prerequisites	NIL	

Course Objectives

1	To learn various software development models.
2	To learn extreme programming and use of GITHUB to manage collaborative software development
3	To Learn simple web page development using HTML.
4	To design interactive web page development using CSS.
5	To understand bootstrap process.

Course Outcomes

Upon completion of the course, students shall have ability to

1	Understand fundamental software development concepts and tools.	[U]
2	Choose methodologies and tools to develop and deploy applications.	[AP]
3	Apply HTML knowledge for static web page development.	[AP]
4	Apply CSS knowledge for designing and implementing dynamic web page.	[AP]
5	Implement web site using java script .	[AP]

Course Contents

Unit I Software Development Model **9 Hrs**

Introduction to software development process, Software engineering principles, requirements gathering and analysis, Overview of programming languages. Agile Software Development: Traditional Software Development Models, Agile methodology (Scrum, Kanban), DevOps practices, Comparison of methodologies.

Case study

Develop a simple application that can perform basic operations using the chosen programming language.

Unit II Extreme Programming and GitHub **9 Hrs**

The Rules of Extreme Programming, Extreme Programming (XP) , Principles, Extreme Programming (XP), Key Terms, Introduction to Lean Software Development, Principles of Lean Software Development, What is Kanban? Introduction to Git , Getting a Git Repository, Recording Changes to the Repository, Viewing the Commit History, Undoing Things, Working with Remotes, Tagging, Git Aliases, Git Branching, Branches in a Nutshell, Basic Branching and Merging, Branch Management, Remote Branches, Rebasing. Introduction to GitHub – Introduction, Set up Git, Create a repository, GitHub Flow, Contribution to Projects, Communicating on GitHub. Linux Basic CommandsX.

Case Study

Imagine a scenario where a team of developers is working on a mobile application. Students will simulate this scenario by collaborating on a shared Git repository to develop a simple mobile app feature. They will create branches for different features, collaborate on code changes, resolve conflicts, and merge their changes back into the main branch. This case study will demonstrate the importance of version control in collaborative

software development.

Unit III HTML **9 Hrs**

Understand the structure of an HTML page, New Semantic Elements in HTML 5, Learn to apply physical/logical character effects, Learn to manage document spacing. Tables - Understand the structure of an HTML table, Learn to control table format like cell spanning, cell spacing, border. List - Numbered List, Bulleted List, Working with Links, Understand the working of hyperlinks in web pages, Learn to create hyperlinks in web pages, Add hyperlinks to list items and table contents. Image Handling - Understand the role of images in web pages, Learn to add images to web pages, Learn to use images as hyperlinks.

Case Study

Building and Engaging Website with HTML Fundamentals.

UNIT IV Introduction to Cascading Style Sheets **9 Hrs**

What CSS can do, CSS Syntax, Types of CSS. Working with Text and Fonts - Text Formatting, TextEffects, Fonts. CSS Selectors Type Selector, Universal Selector, ID Selector, Class selector. Colors and Borders Background, Multiple Background, Colors RGB and RGBA, HSL and HSLA, Borders, Rounded Corners, Applying Shadows in border, Implementing CSS3 in the "Real World" Modernizr, HTML5 Shims, SASS, and Other CSS Preprocessors, CSS Grid Systems, CSS Frameworks.

Case Study

Consider a scenario where a software company is tasked with maintaining an existing codebase that lacks proper documentation and has accumulated technical debt over time. Students will analyze the codebase, identify areas for improvement, and refactor the code to adhere to clean code principles. They will then write unit tests to ensure that the refactored code behaves as expected, demonstrating the importance of clean code and testing in software maintenance.

Unit V Bootstrap **9 Hrs**

Introduction, Getting Started with Bootstrap, Bootstrap Basics, Bootstrap grid system, Bootstrap Basic Components, Bootstrap Components, Page Header, Breadcrumb, Button Groups, Dropdown, Nav & Navbars. JavaScript Essentials - Var, Let and Const keyword, Arrow functions, default arguments, Template Strings, String methods, Object de-structuring, Create, apply, prototype, bind method, Spread and Rest operator, Typescript Fundamentals, Types & type assertions, Creating custom object types, function types, Typescript OOPS - Classes, Interfaces, Constructor, Decorator & Spread Operators Race.

Case Study

Imagine a scenario where a team of developers is working on a web application that needs to be deployed to a cloud platform. Students will set up a CI/CD pipeline using tools like Jenkins or GitHub Actions to automate the build, test, and deployment process. They will deploy the application to a cloud platform (e.g., AWS, Heroku) and configure the pipeline to trigger automatic deployments whenever new changes are pushed to the repository. This case study will showcase the benefits of CI/CD in streamlining the software development lifecycle.

Total Hours	**45**

Laboratory Component

S. No	List of Experiments
1	Gather application specific requirements for assimilate into RE (Requirements engineering) model.
2	Select relevant process model to define activities and related tasks set for assigned project.
3	Set up a Git repository and perform basic version control operations (commit, push, pull, merge).

4	Collaboratively work on a project using Git, branching, and merging.
5	Perform manual testing of a small software application and document test cases and results.
6	Write unit tests for a simple function or module using a testing framework (e.g., JUnit for Java, unittest for Python).
7	Refactor a piece of code to improve its readability, following clean code principles.
8	Work on an agile project using Scrum or Kanban, with iterative development sprints.
9	Set up a CI/CD pipeline using a tool like Jenkins or GitHub Actions for a sample application.
10	Deploy a web application to a cloud platform (e.g., AWS, Heroku) using CI/CD pipelines.
Project Work	
	Collaboratively develop a small software application using version control, testing, and deployment practices learned during the course. Present the project to the class, demonstrating key features, code quality, and deployment process.
Text Books	
1	Software Engineering: A Practitioner's Approach" by Roger S. Pressman (2023).
2	Jeff Sutherland, “Scrum the Art of Doing Twice the Work in Half the Time”, Random House Publisher, 1st Edition, (2022).
3	Scott Chacon, Ben Straub, “Pro GIT”, Apress Publisher, 3rd Edition, (2024).
4	Code Complete: A Practical Handbook of Software Construction" by Steve McConnell (2014).
5	Refactoring: Improving the Design of Existing Code" by Martin Fowler (2018).
6	The Art of Software Testing" by Glenford J. Myers, Corey Sandler, and Tom Badgett (2011).
7	Continuous Integration: Improving Software Quality and Reducing Risk" by Paul M. Duvall, Steve Matyas, and Andrew Glover (2007).
Reference Books	
1.	Robert C. Martin, “Agile Software Development, Principles, Patterns and Practices”, Prentice Hall, 2nd Edition, (2020).
2.	Mike Cohn, “User Stories Applied: For Agile Software”, Addison Wesley, 2nd Edition, (2016).
3.	Software Engineering: A Practitioner's Approach" by Roger S. Pressman, (2014).
4.	Scrum: The Art of Doing Twice the Work in Half the Time" by Jeff Sutherland (2014)
5.	Continuous Delivery: Reliable Software Releases through Build, Test, and Deployment Automation" by Jez Humble and David Farley (2023).
6.	Pro Git" by Scott Chacon and Ben Straub (2023).
7.	Version Control with Git" by Jon Loeliger and Matthew McCullough (2020).
8.	Jenkins: The Definitive Guide" by John Ferguson Smart (2020).
9.	The Phoenix Project: A Novel About IT, DevOps, and Helping Your Business Win" by Gene Kim, Kevin Behr, and George Spafford (2013)
Web References	
1.	https://www.coursera.org/specializations/agile-development

2.	https://www.edx.org/learn/agile
3.	https://nptel.ac.in/courses/106/105/106105182/
4.	https://www.agilealliance.org/
5.	https://www.scrum.org/
6.	https://trello.com/
Online Resources	
1.	http://www.agilenutshell.com/
2.	https://www.atlassian.com/agile/scrum
3.	https://www.youtube.com/user/AgileMikeCohn
4.	https://guides.github.com/
5.	https://git-scm.com/doc

24UCSES102	Programming and Problem Solving using C++	3/0/2
Nature of Course	Theory & Programming	
Prerequisites	NIL	

Course Objectives

1	To learn the fundamental concepts and methodologies for C++ programming.
2	To provide the basic object-oriented programming concepts using C++.
3	To learn reusability concept using C++ inheritance.
4	To learn reusability concept using C++ polymorphism.
5	To introduce file streams and operations for storing data permanently.

Course Outcomes:
Upon completion of the course, students shall have ability to

1	**Understand** the fundamental concepts and methodologies required to develop a C++ program for given problems.	[U]
2	**Develop** a program for real life problems using object oriented concepts.	[A]
3	**Apply** the inheritance concept to solve real life problems.	[A]
4	**Develop** C++ programs using reusability concept.	[A]
5	**Implement** the concepts on file streams, I/O and Lambda Expression.	[A]

Course Contents

Unit I Programming Fundamentals in C+ **9 Hrs**

C++ Fundamentals: Header files, Basic Input and Output (I/O) usingcin and cout, Variable, Syntax, Data-type, variables, strings **Operators:** Arithmetic Operators, Assignment Operators, Relational Operators, Logical Operators, Bitwise Operators, Other Operators, and Operator Precedence. **Control Statements**: if, if...else and Nested if...else, switch. Case, break and continue, **Loops** for loop, while loop, do while loop, goto.

Case Study

Develop an C++ Application for Banking , Bank management system that allows banks to manage and organize their customer accounts efficiently. It simplifies various banking operations such as account creation, deposit, withdrawal.

Unit II Object Oriented Concepts **9 Hrs**

Classes and Objects, public, private, protected. Constructors and destructors: Overloaded Constructor, Copy Constructor, and Shallow Copying Deep Copying. Member data, member functions, inline function, friend functions, static members, and references.

Case Study

Develop an C++ application using constructor for Students Grade: Design a C++ class named studentgrade to represent a graduating student. The class should encapsulate the following information about a student:
Student ID: A unique identifier (e.g., integer) for the student.
Name: The student's full name (e.g., string).
GPA: The student's cumulative grade point average (e.g., double).

Unit III Inheritance **9 Hrs**

Inheritance: Class hierarchy, derived classes, single inheritance, multiple, multilevel, hybrid inheritance, role of virtual base class, constructor and destructor execution, base initialization using derived class constructors.

Case Study

Develop a C++ application for employee salary calculation: Create a class called Employee that has private member variables for name, employee ID, and salary. Include member functions to calculate and set salary based on employee performance.

Unit IV Polymorphism **9 Hrs**

Polymorphism: Binding, Static binding, Dynamic binding, Static polymorphism: Function Overloading, Ambiguity in function overloading, Operator Overloading Dynamic polymorphism: Base class pointer, object slicing, late binding, method overriding with virtual functions, pure virtual functions, abstract classes.

Case Study

Develop an C++ application for Hotel management system: Design and implement a hotel management system in C++. The system should allow hotel staff to manage room bookings, guest information, check-in/check-out processes, and room availability.

Unit V Files and Generic Programming **9 Hrs**

Abstract Classes as Interfaces, Exception, Files, Streams and I/O, STL, Generic Programming, Lambda Expression.

Case Study

Develop C++ application for file manipulation: Develop a C++ program that reads data from an input file, performs specific operations or calculations on the data, and then writes the results to an output file.

Total Hours	**45 Hrs**

Lab Component

S.No.	**Lab Exercise**
1.	Write a C++ program to print the integers in the reverse order. □1 <= N <= 1000 □1 <= a[i] <= 10000, where a[i] is the ith integer in the array.
2.	Create a class named students with the following specifications: a) □An instance variable named scores to hold a student's 5 exam scores. b) □A void input() function that reads 5 integers and saves them to scores. c) □An int calculateTotalScore() function that returns the sum of the student's scores.
3.	Create a class named 'Rectangle' with two data members- length and breadth and a function to calculate the area which is 'length*breadth'. The class has three constructors which are : 1 - having no parameter - values of both length and breadth are assigned zero. 2 - having two numbers as parameters - the two numbers are assigned as length and breadth respectively. 3 - having one number as parameter - both length and breadth are assigned that number. Now, create objects of the 'Rectangle' class having none, one and two parameters and print their areas.
4.	Create three classes Person, Professor and Student. The class Person should have data members name and age. The classes Professor and Student should inherit from the class Person. The class Professor should have two integer members: publications and cur_id. There will be two member functions: getdata and putdata. The function getdata should get the input from the user: the name, age and publications of the professor. The function putdata should print the name, age, publications and the cur_id of the professor. The class Student should have two data members: marks, which is an array of size 6 and cur_id. It has two member functions: getdata and putdata. The function getdata should get the input from the user: the name, age, and the marks of the student in 6 subjects. The function putdata should print the name, age, sum of the marks and the cur_id of the student. For each object being created of the Professor or the Student class, sequential id's should be assigned to them starting from 1. Solve this problem using virtual functions, constructors and static variables. You can create more data members if you want.
5.	Write a program to design a class representing complex numbers and having functionality of performing addition and multiplication of two complex numbers using operator overloading.
6.	Create two classes HotelRoom and HotelApartment denoting respectively a standard hotel room and a hotel apartment. An instance of any of these classes has two parameters: bedrooms and

	bathrooms denoting respectively the number of bedrooms and the number of bathrooms in the room. The prices of a standard hotel room and hotel apartment are given as: Hotel Room: 50 bedrooms + 100 bathrooms. Hotel Apartment: The price of a standard room with the same number bedrooms and bathrooms plus . For example, if a standard room costs 200, then an apartment with the same number of bedrooms and bathrooms costs 300. In hotel's codebase, there is a piece of code reading the list of rooms booked for today and calculates the total profit for the hotel. However, sometimes calculated profits are lower than they should be.
7.	Implement a program to handle exception it prints appropriate error messages, should anything go wrong. The expected behavior is defined as follows: If the compute function runs fine with the given arguments, then print the result of the function call. If it fails to allocate the memory that it needs, print Not enough memory. If any other standard C++ exception occurs, print Exception: S where S is the exception's error message. If any non-standard exception occurs, print Other Exception.
8.	Write a C++ Program to implement input and output operation on file
9.	You are appointed as the assistant to a teacher in a school and she is correcting the answer sheets of the students. Each student can have multiple answer sheets. So the teacher has Q queries: 1 X Y:Add the marks Y to the student whose name is X. 2 X: Erase the marks of the students whose name is X. 3 X: Print the marks of the students whose name is X. (If X didn't get any marks print 0.)
10.	Mini Project

Total Hours (Lab)	**30**
Total Hours(45+30)	**75**

Text Books:

1.	E Balagurusamy,"Object Oriented Programming With C++", 4th Edition, Tata McGraw-Hill Education, (2020).
2.	YashavantP. Kanetkar, "Let us C++", BPB Publications, (2020).
3.	M. Sprankle, "Problem Solving and Programming Concepts", 9th Edition, Pearson Education, New Delhi, (2011).

Reference Books:

1.	Herbert Schildt, "The Complete Reference C++", 4th edition,MH, (2015).
2.	John Hubbard, "Schaum's Outline of Programming with C++", MH, (2016).

Web References:

1.	https://www.geeksforgeeks.org/c-plus-plus/
2.	http://web.stanford.edu/class/cs106l/

Online Resources:

1.	https://nptel.ac.in/courses/106101208
2.	https://www.hackerrank.com/domains/cpp
3.	https://codeforces.com/blog/entry/74684
4.	https://www.hackerearth.com/practice/notes/tricky-and-fun-programming-in-c/

24UETAE101	English - Oral and Written Communication Skills	1/0/2
Nature of Course	Theory	
Prerequisites	NIL	
Course Objectives		
1	To identify and list various reading techniques, types of formal correspondence, speaking strategies, listening exercises, and essential grammar and vocabulary components.	
2	To summarize the importance of reading, writing, speaking, and listening skills and their applications in workplace scenarios.	
3	To apply reading strategies for comprehension, write formal emails and reports, engage in professional conversations, and use listening skills to understand and respond to different audio materials.	
4	To analyze case studies to identify effective communication strategies and areas for improvement in professional settings.	
5	To evaluate their own and others' communication practices through role plays, writing assignments, and listening exercises, providing constructive feedback.	
6	To create well-structured resumes, job application letters, technical reports, and proposals, and design effective communication strategies for professional use.	
Course Outcomes		
1	Students will be able to list different reading techniques, types of formal correspondence, speaking strategies, listening exercises, and essential grammar and vocabulary components.	[U]
2	Students will be able to summarize the importance of reading, writing, speaking, and listening skills and their applications in workplace scenarios.	[U]
3	Students will be able to apply reading strategies for comprehension, write formal emails and reports, engage in professional conversations, and use listening skills to understand and respond to different audio materials.	[AP]
4	Students will be able to analyze case studies to identify effective communication strategies and areas for improvement in professional settings.	[A]
5	Students will be able to evaluate their own and others' communication practices through role plays, writing assignments, and listening exercises, providing constructive feedback.	[AP]
6	Students will be able to create well-structured resumes, job application letters, technical reports, and proposals, and design effective communication strategies for professional use.	[AP]
Course Contents		
UNIT I Advanced Reading and Comprehension		**3 Hrs**
Reading Techniques: Skimming, scanning, and cloze reading, Technical Reading: Understanding technical articles and reading for detailed comprehension, including emails, letters, and advertisements, Data Interpretation, Interpreting charts and graphs, table completion, Verbal Reasoning: Comprehending reviews and		

instant messages, enhancing verbal reasoning skills.

Case Study

Analyzing the Impact of Skimming, Scanning, and Technical Reading in an Engineering Firm.

UNIT II Professional Writing Skills **3 Hrs**

Formal Correspondence: Crafting formal letters (sales, quotations, clarification, orders, complaints, invitations), Email Writing, Effective email communication. Documentation, Writing minutes of meetings, professional reports, and proposals, Career Documents: Developing resumes and job application letters. Analyzing and documenting case studies.

Case Study

Enhancing Professional Communication at GlobalCorp

UNIT III Speaking Skills **3 Hrs**

Conversational Speaking: Initiating, maintaining, and closing conversations. Deciphering Pictures and Mind Maps: Using pictures and mind maps as conversation starters; discussing and interpreting visual information. Answering Questions: Responding to questions based on situational contexts; focusing on clarity, coherence, and appropriateness of responses. Interactive Speaking: Pronunciation challenges, intonation, stress, and rhythm. Role Plays and Extempore: Engaging in role plays and extempore speaking; organizing thoughts quickly and speaking confidently.

Case Study

Improving Conversational and Interactive Speaking Skills at EduTech.

UNIT IV Listening Skills **3 Hrs**

Situational Listening: Listening to various situational recordings, focusing on understanding the context and main ideas. Listening to Experiences: Hearing personal stories and experiences, practicing summarizing and discussing the content. Short Extracts: Listening to short audio or video extracts, identifying key points and details. Interviews and TV News: Listening to interviews and TV news segments, practicing note-taking and extracting relevant information. Interactive Listening: Focusing on techniques such as summarizing, paraphrasing, and asking clarifying questions, practicing listening for specific details in different types of audio materials, engaging in tasks that require identifying key information.

Case Study

Enhancing Listening Skills in Customer Support at ServicePlus.

UNIT V Integrative Grammar and Vocabulary Development **3 Hrs**

Grammar Essentials: Parts of speech, verbs, modals, tenses, sentence formation, homonyms, homographs, and homophones, Vocabulary Building: Error analysis, synonyms and antonyms, word formation, collocations, technical vocabulary, analogies, and business vocabulary, Writing Techniques: Note-making, précis writing, essay writing, technical report writing, memo writing, letter writing, and email writing, Listening and Speaking: Distinguishing native and Indian English, listening to TED talks, podcasts, and speeches by prominent figures, engaging in classroom activities, and telephonic enquiries.

Case Study

Boosting Grammar and Vocabulary Proficiency at MediaWorks.

	Total Hours	**15**
Lab Content	**Total Hours**	**30**

1. Students compete to quickly identify key information in a text using skimming and scanning techniques. This activity enhances speed reading and efficient information retrieval skills.

2. Students rotate through stations, each offering different reading activities and challenges. This promotes engagement and diverse reading skills development.

3. Participants dissect technical articles to comprehend complex information and specialized vocabulary. This improves proficiency in reading technical texts.

4. Learners practice interpreting data from charts, graphs, and tables. This activity enhances their ability to understand and analyze quantitative information.

5. Students apply their reading and comprehension skills to real-world situations. This connects theoretical knowledge with practical use.

6. Participants examine texts to determine the author's tone and its impact on the message. This develops sensitivity to nuance and intent in writing.

7. Students act out scenarios to practice proper etiquette in different contexts. This enhances social and professional interaction skills.

8. Learners' draft and respond to emails in simulated professional settings. This improves their email etiquette and communication efficiency.

9. Students practice initiating and sustaining conversations on various topics. This builds confidence and fluency in speaking.

10. Participants interpret visual media and discuss their insights. This develops skills in analyzing and articulating visual information.

11. Students create mind maps to organize and connect ideas from texts. This fosters creativity and deeper understanding of material.

12. Learners take turns telling stories, focusing on narrative structure and engagement. This boosts their storytelling and public speaking skills.

13. Participants write journal entries reflecting on their learning experiences. This encourages self-awareness and critical thinking.

14. Students create timelines using different verb tenses to describe sequences of events. This clarifies their understanding of tense usage.

15. Students practice using specialized vocabulary in simulated business or technical scenarios. This boosts their proficiency in professional language.

Text Books:

1	Technical Communication: English Skills for Engineers by Meenakshi Raman and Sangeeta Sharma (2008), Oxford University Press
2	Communication Skills by Sanjay Kumar & Pushp Lata (2018), Oxford University Press
3	Creative English for Communication by N. Krishnaswamy and T. Sriraman (3rd Edition), Macmillan India Limited.
4	A Course in Technical English by Dr.Praveen Sam and K N Shoba (2020), Cambridge University Press.

Reference Books:

1	Business Correspondence and Report Writing by RC Sharma and Krishna Mohan (2017), McGraw Hill Education
2	English Grammar, Composition and Usage by NK Aggarwal and FT Wood (4th Edition), Macmillan Publishers India
3	Business Communication by S.M. Rai and Urmila Rai (2015), Himalaya Publishing House
4	Longman English Grammar Practice for Intermediate Self Study Edition with Key by L.G. Alexander (2022), Longman
Web References:	
1	http://www.academiccourses.com/Courses/English/Business-English
2	https://www.bbc.co.uk/learningenglish/basic-vocabulary
3	https://www.teachingenglish.org.uk/professional-development/teachers/knowing-subject/articles/using-intonation
4	https://www.thebalancemoney.com/cover-letters-6265516
Online Resources:	
1	https://www.coursera.org/specializations/business-english
2	https://www.businessenglishresources.com/learn-english-for-business/student-section/practice-exercises-new/
3	https://www.esl-lab.com/
4	https://breakingnewsenglish.com/

24UETVS101	Design Thinking and Idea Lab	0/0/4
Nature of Course	Programming	
Prerequisites	NIL	
Course Objectives		
1	Enhance teamwork and problem-solving skills for real-world challenges.	
2	Develop solutions through value proposition, assumptions, and user focus.	
3	Apply empathy and data analysis to inform creative thinking.	
4	Generate and refine innovative solutions through established methods.	
5	Prototype solutions using appropriate tools.	
Course Outcomes		
1	Prepare a team and utilize problem solving skills for real world challenges.	[AP]
2	Implement solutions through value proposition, assumptions, and user focus.	[AP]
3	Apply empathy and data analysis to inform creative thinking.	[AP]
4	Analyse and refine innovative solutions through established methods.	[A]
5	Propose prototype solutions using appropriate tools.	[A]

Laboratory Component	
S.No	**List of Experiments**
1.	Team building and problem statement identification (Clearly mention the problem your group would like to solve) a) Mission Statement (Why is it important to solve this problem? Who will be the beneficiaries? What is the market opportunity?) b) Value Proposition (Clearly state the redefined problem with specific issues the team would like to solve)
2.	Assumptions (What are the current/existing considerations/limitations regarding the problem your team would like to address?) a) Stakeholders (List all the stakeholder groups that can influence or can be influenced by a change. Which stakeholder group(s) will be benefitted? Which stakeholder group(s) has your team interacted with? – Identify which user group you would like to target the solution - Mainstream, Extreme or Latent users)
3.	Empathy Tool Used (What/How/Why, Empathy Map, AEIOU method, Beginner's mindset, Story/Share capture, etc.)

4.	Data Collection (Research, Questionnaires, Interviews, Surveys, Stakeholder groups, Statistics, etc.)
5.	Insights (Document all points from data collection stage to form insights about the problem)
6.	Ideation Method Used (Mind Map, Brainstorming, SIT method, SCAMPER, Three-Box Thinking)
7.	Refine innovation ideas using design heuristics. Apply research-based personas and behavior models to make innovations easier to adopt.
8.	Design project prototype using pop marvel software.
Text Books:	
1.	Brown, Tim, "Change by Design: How Design Thinking Creates New Alternatives for Business and Society," 1st edition, Harper Business.
2.	Liedtka, Jeanne, and Tim Ogilvie, "Designing for Growth: A Design Thinking Tool Kit for Managers," 1st edition, Columbia Business School Publishing.
3.	Kelley, Tom, and David Kelley, "Creative Confidence: Unleashing the Creative Potential Within Us All," 1st edition, Crown Business.
Reference Books	
1.	Martin, Roger L., "The Design of Business: Why Design Thinking is the Next Competitive Advantage," 1st edition, Harvard Business Review Press.
Web References	
1.	https://dschool.stanford.edu/resources/design-thinking-bootleg
2.	https://www.ideou.com/
Online Resources	
1.	https://www.coursera.org/courses?query=design%20thinking

SEMESTER – II

24UCSBS102	Engineering Mathematics-II	3/1/0
Nature of Course	Theory	
Prerequisites	Engineering Mathematics- I	
Course Objectives		
1	To apply numerical techniques to solve engineering problems and evaluate the results.	
2	To analyse and solve ordinary differential equations by computational methods.	
3	Study Linear Programming Problems , transportation problems and various methods of solving them.	
4	To study the techniques of complex variables and functions together with contour integrations.	
5	To be able to calculate area and volume by using double and triple integrals.	
Course Outcomes (Theory) **Upon completion of the course, students shall have ability to**		
1	Calculate value of definite integrals numerically with the help of numerical methods.	[AP]
2	Determine solution of differential equations with the help of numerical methods.	[AP]
3	Solving LPPs by graphical and simplex method and finding the optimal solution to TP.	[AP]
4	Compute several kinds of real integrals using the residue theorem and Cauchy's theorem.	[AP]
5	Determine area, volume using double and triple integration.	[AP]
Course Contents		
Each unit will contain real time applications using geogebra.		

UNIT I Numerical Methods-I **9 Hrs**

Interpolation- Forward, backward, Lagrange's; Newton's Divided Difference Formulae; Numerical Integration- Trapezoidal, Simpson's 1/3,3/8.

Case Study

Demonstrate forward and backward interpolation methods Using Geogebra to demonstrate forward and backward interpolation methods. Allow students to input a set of data points and choose the desired order of interpolation. Visualize how the interpolated polynomial passes through the given points and how changing the order affects the accuracy of the interpolation.

UNIT II Numerical Methods-II **9 Hrs**

Numerical solution of Ordinary Differential Equations Introduction, Picard's Method, Euler's method, Modified Euler's method, Runge-Kutta method.

Case Study

Numerical Solution of ODEs: Using Geogebra to introduce the concept of numerical solution methods for ordinary differential equations (ODEs). Provide an overview of Picard's method, Euler's method, Modified Euler's method, and Runge-Kutta methods. Illustrate the motivation behind numerical methods and their applications in solving ODEs.

UNIT III Finite Fields **9 Hrs**

Group, Abelian group, Ring, Commutative ring, Integral domain, Field, the set Zn and its properties, Prime numbers, Fermat's Little Theorem, Euler's Theorem, Congruences and its properties, Euclidean Algorithm for finding gcd, Arithmetic operations on polynomials whose coefficients are from finite fields.

Case Study

Visualization of RSA Encryption and Decryption: Develop a Geogebra applet to visualize RSA encryption and decryption operations. Allow students to input values for the public and private keys (n, e, d), and a plaintext message. Visualize how the message is encrypted using the public key and decrypted using the private key, highlighting the modular exponentiation operations involved.

UNIT IV Complex Analysis **9 Hrs**

Analytic and Harmonic functions, Integration, Cauchy's Theorem, Singularities, Calculus of Residues, Applications to Evaluate Real Integrals.

Case Study

Integration in Complex Analysis: Using Geogebra demonstrations to illustrate integration in complex analysis. Allow students to input complex functions and contours, and visualize how contour integration is used to evaluate complex integrals.

UNIT V Multiple Integrals **9 Hrs**

Double integrals in cartesian and polar coordinates, change of order, triple integrals in cartesian, spherical and cylindrical coordinates, applications to area, volume.

Case Study

Applications to Area and Volume: Geogebra demonstrations to illustrate applications of double and triple integrals in computing areas and volumes. Allow students to input functions representing surfaces and visualize how double and triple integrals can be used to compute surface areas and volumes of solids. Examples of real-world applications in engineering and physics.

Total Hours:	**45**

Text Books	
1	Introductory Methods of Numerical Analysis, S. S. Sastry, 5th edition (2012).
2	H. K. Das, Advanced Engineering Mathematics, S Chand, (2019).
3	Modern Algebra, A. R. Vasishtha, A. K. Vasishtha, ISBN: 9789387620414 Edition: (2023).
Web References	
1	https://onlinecourses.nptel.ac.in/noc23_ma44/preview
2	https://onlinecourses.nptel.ac.in/noc23_ma51/preview
3	https://onlinecourses.swayam2.ac.in/cec23_ma02/preview

4	https://engineering.purdue.edu/kak/compsec/
Reference Books	
1	R. K. Jain and S. R. K. Iyengar, Advanced Engineering Mathematics, Narosa Publishing House, 2014, ISBN-13: 978-1842653418.
2	N. P. Bali and Manish Goyal, A TextBook of Engineering, Mathematics, 8/e, Lakshmi Publications, 2016, ISBN: 8131808327
3	Erwin Kreyszig, Advanced Engineering Mathematics, Wiley, 9/e, 2016, ISBN-13: 978-0471488859.
4	E.B. Staff & A.D. Snider, Fundamentals of Complex Analysis with Application to Engineering and Science 3rd edition, , ISBN: 0139078746
Online Resources	
1	https://youtube.com/playlist?list=PLyqSpQzTE6M-QT7PvEBHV0iNMvZk9mocO&feature=shared
2	https://youtube.com/playlist?list=PLbMVogVj5nJTLfYTwvct_SlLaxv1b50Vk&feature=shared
3	https://youtube.com/playlist?list=PLtKWB-wrvn4nA2h8TFxzWL2zy8O9th_fy&feature=shared

24UETBS103	APPLIED SCIENCE	4/0/0
Nature of Course	Theory	
Prerequisites	Higher Secondary Basic Science	
Course Objectives		
1	To learn and comprehend the essentials of physics and its application in computer science.	
2	To study the electrical properties and semiconductor physics of materials and apply in the engineering field.	
3	To understand the basic parameters involved in water technology and water treatment methods for domestic use.	
4	To learn the cause and management of electronic waste.	
5	To study the basics of quantum computing, nanomaterials and applications.	
Course Outcomes (Theory) **Upon completion of the course, students shall have ability to**		
1	Utilize the concepts of waves, semiconductor and ancient astronomical techniques in the field of engineering.	[AP]
2	Implement the knowledge of LASERS and fibre optics in engineering field.	[AP]
3	Execute the acquired ideas on water treatment and analysis.	[AP]
4	Apply the gained knowledge in various environmental issues in sustainable development.	[AP]
5	Conceptualise the basics of quantum computing and nanoscience. Implement the concepts of physics in animation.	[U]
Course Contents		

UNIT I Introduction to Fundamental Physics **12 Hrs**

Waves and Oscillation: Types of Waves, Standing Waves (Nodes and Antinodes), Superposition of Waves, and Its Applications. SHM: Wave Equations, Simple Pendulum (Problems). Electrodynamics: Faraday's Laws, Lenz's Law, Biot-Savart Law, Physics of Divergence, Gradient, and Curl, Magnetic Field, Vector Potential, Maxwell Equations (Qualitative). Semiconductor Electronics: Band theory of solids, Energy band diagram, Semiconductor: Properties, Type, P-N Junction diode and applications, I-V characteristics, P-N-P and N-P-N transistors, Modern Physics: Photoelectric Effect, Einstein's Photoelectric equation, Photocell(working), Solar cell. Ancient Astronomical Instruments: Chakra, Dhanuryantra, Yasti and Phalak Yantra etc.

Case Study
Simulate the behaviour of a simple electronic circuit using a circuit simulator
Prepare a presentation on a given topic and present in group.
Demonstrate a given concept through a model.
Scenario: Apply concepts of waves and oscillation, electrodynamics, and semiconductor electronics to a real-world scenario

UNIT II Optics and Introduction to Quantum Physics **12 Hrs**

Light: Properties of Light, TIR Critical Angle, Classification of Optical Fibres, Attenuation and Fibre Losses, Light propagation through Fibres, Acceptance Angle and Numerical Aperture (NA), Applications Fibre Optic Communication, Block Diagram. Laser Characteristics, Spatial and Temporal Coherence, Einstein Coefficient & Expression for Energy Density, Energy Level Diagram, Pumping Schemes, Components of Laser, Nd-YAG, He-Ne, and Dye Laser and Its Applications. Sensors: Introduction, Hall Effect and Hall Sensor, Applications of Sensors. Quantum Physics: de'Broglie Hypothesis and Matter Waves, de'Broglie Wavelength, Phase Velocity and Group Velocity, Heisenberg's Uncertainty Principle and Its Application.

Case Study
Demonstrate given physical phenomenon through experiment.Video record it and share url.
Prepare presentations on given topics and present in groups.
Simulate the given experimental setup by using Virtual Labs/XCos
Scenario: Apply concepts of light, laser, and sensors to a real-world scenario, and understand the principles of quantum physics.

UNIT III Water Technology **12 Hrs**

Water Technology: Introduction; Hardness of Water- Cause, Types, Units, Disadvantages of Using Hard Water for Domestic and Industrial Purposes (Scale and Sludge Formation in Boilers, Caustic Embrittlement, Boiler Corrosion), Softening of Hard Water (Lime-Soda, Permutit and Ion-Exchange Processes); Chemical Survey of Water- Analysis of Free Chlorine, Total Alkalinity, Hardness and Dissolved Oxygen, Biological Oxygen Demand, Numerical Based on Determination of Hardness, Internal Treatment Methods-Specifications for Drinking Water, BIS & WHO Standards, Treatment of Water for Domestic Use, Desalination - Reverse Osmosis & Electrodialysis.

Case Study
Conduct a survey (by using internet) on the water quality standards for groundwater found in various parts of India, and evaluate the criteria based on WHO/BIS guidelines.
Scenario: Apply water quality analysis concepts to achieve the required survey results.

UNIT IV Management of Electronic Waste **12 Hrs**

Management of Electronic Waste: Overview of Electronic Waste, Sources of Electronic Waste, Various Types of Electronic Waste, Hazardous Materials Utilised in The Production of Electrical and Electronic Goods; Environmental Impact of Electronic Waste, Adverse Effect of Electronic Waste on Human Health, The Necessity of Electronic Waste Management, Recycling and Recovery of Electronic Waste: Various Methods of Recycling (Separation, Thermal Treatments), Hydrometallurgical Extraction, Pyrometallurgical Methods and Direct Recycling. Extraction of Nanogold from Electronic Trash. Participant Role in Electronic Waste Management.

Case Study
Conduct a survey (by using internet) on different types of electronic wastes and their recycling/ recovery processes.
Scenario: Ensure that the process of recycling and recovery is sustainable.

UNIT V Basics of Quantum Computing, Animation Physics and Nanotechnology 12 Hrs

Basics of Quantum Computing: Introduction to Quantum Computing, Differences Between Classical & Quantum Computing, Representation of qubit by Bloch sphere, Single qubit gates, Matrix representation of |0> and |1> States and their multiplication (Inner Product), Normalisation rule, Numerical Problems.

Physics of Animation: Frames per Second, Size and Scale, Motion and Timing in Animations, The Odd rule, Motion Graphs, Examples of Character Animation: (Jumping, Walking: Strides and Steps, Numerical Problems.)

Nanotechnology Introduction to Nanomaterials, Nanoparticles, Nanoclusters, Carbon Nanotubes (CNTs) and Nanowires. Synthetic Techniques of Nanomaterials, Characterization of Nanomaterials via Scanning Electron Microscope (SEM) and Transmission Electron Microscope (TEM). Application of Nanomaterials (Biology, Medicine, IT and Electronics)

Case Study

Analysis of Physics-Based Animation in a Real-World Movie or Game. Interactive 3D models of animations can be created to demonstrate the concept of motion and timing in animations.

Investigate the nanotechnology associated with nature.

Scenario - Apply concepts of nanotechnology, physics of animation to a real-world scenario. Understand the principles of motion, energy, and forces in animation. Analyse the role of physics in creating realistic animations. Develop critical thinking and problem-solving skills

Total Hours	**60**

Text Books

1	A Textbook of Engineering Physics, M N Avadhanulu, S Chand & Company Pvt. Ltd., (2014), ISBN: 978-8-121-90817-7.
2	Quantum Computation and Quantum Information, Michael A. Nielsen & Isaac L. Chuang, Cambridge Universities Press, (2010) ISBN: 978-0-511-97666-7.
3	Physics for Animators, Michele Bousquet with Alejandro Garcia, CRC Press, Taylor & Francis, 2016. ISBN: 978-0-415-84297-6.
4	The Surya Siddhanta by Aryabhatta Baptist mission press, Calcutta. (1861).
5	Engineering Chemistry: Fundamentals and Applications, Shikha Agarwal, Cambridge University Press, (2016) ISBN: 978-1-107-47641-7.
6	Handbook of Electronic Waste Management, Majeti Narasimha Vara Prasad, Meththika Vithanage, Anwesha Borthakur, Elsevier Publisher, 2019, ISBN: 978-0-128-17030-4
7	A textbook on Nanoscience and Nanotechnology, T. Pradeep, Tata McGraw Hill Education Private Limited, (2012), ISBN: 978-0-071-54829-8

Reference Books

1	Principles of Engineering Physics, Cambridge University Press, 2018, ISBN:978-1-316-87694-7
2	Engineering Physics by Gupta and Gour, Dhanpat Rai Publications, 2016 (Reprint). ISBN:978-8-189-92822-3

3	Lasers and Non Linear Optics, B B Loud, New age international, 2011 edition. ISBN:978-8-122-40324-4
4	Concepts of Modern Physics, Aurthur Beiser, McGrawhill, 6th Edition, 2009, ISBN:978-0-071-00144-1
5	Information, Physics and Computation (Oxford Graduate Texts), 2009, M. Mézard, A. Montanari, ISBN:978-0-198-57083-7
6	Engineering Chemistry, K. Sesha Maheswaramma, Mridula Chugh, Pearson Education India, 2015, ISBN:978-8-131-77451-9
7	Engineering Chemistry, Satyaprakash & Manisha Agrawal, Khanna Book Publishing, 2012, ISBN-13. 978-9380016757
8	An Introduction to Nanomaterials and Nanoscience, Asim K. Das, Madhu Das, CBS Publishers, 2022, ISBN: 978-9-385-91567-3
9	Engineering Chemistry, C. Baskar, S. Baskar, R.S. Dhillon, Wiley, 2012, ISBN: 978-8126532018
Web References	
1	NPTEL Quantum Computing: https://archive.nptel.ac.in/courses/115/101/115101092
2	Physics Simulation: https://phet.colorado.edu/sims/html/plinko-probability/latest/plinkoprobability_en.html
3	Physics of Animation: https://www.youtube.com/watch?v=kj1kaA_8Fu4
4	https://ocw.mit.edu
5	NPTEL Nanomaterials https://www.youtube.com/playlist?list=PLyqSpQzTE6M8682dGkNTN8936vSY4CbqZ
6	NPTEL Electronic Waste Management: https://www.youtube.com/playlist?list=PLQC-_J4xl9DT0LuLJAKspLoc8Xq2IWYzb
7	NPTEL Water Technology https://www.youtube.com/playlist?list=PLLy_2iUCG87AZvtaiuD3r4HATrBKhb90P
Online Resources	
1	Virtual LAB: https://www.vlab.co.in/participating-institute-amrita-vishwa-vidyapeetham
2	Virtual LAB: https://vlab.amrita.edu/index.php?sub=1&brch=189&sim=343&cnt=1
3	Virtual LAB: https://virtuallabs.merlot.org/vl_physics.html
4	Virtual LAB: https://ee1-nitk.vlabs.ac.in/List%20of%20experiments.html
5	Modelling & Simulation: https://cloud.scilab.in/
6	Modelling & Simulation: https://www.compadre.org/osp/

7	Modelling & Simulation: https://www.circuitlab.com/editor/#?id=7pq5wm&from=homepage

24UETBS104	**APPLIED SCIENCE LAB**	**0/0/4**
Nature of Course	Programming	
Prerequisites	10+2 Basic Science	
Course Objectives		
1	To make the students gain practical knowledge to correlate with theoretical studies.	
2	To achieve perfectness in experimental skills and to bring more confidence and ability in developing and fabricating engineering and technical equipment.	
3	To apply ethical principles and to commit to professional ethics and responsibilities.	
4	To develop communication skills and to discuss the basic principles of Engineering Physics and Chemistry in a group.	
Course Outcomes (Theory) **Upon completion of the course, students shall have ability to**		
1	Utilize working knowledge of fundamental Physics and basic electrical and or mechanical engineering principles to evaluate.	[AP]
2	Utilise working knowledge of fundamental Chemistry in engineering applications and technology.	[AP]
3	Perform experiments skill fully, record the data with more precision, interpret results and communicate effectively.	[AP]
4	Report the experimental work in stipulated time by following ethical principles.	[AP]

Lab Components	
1	Determine Energy Band Gap of Semiconductor
2	Effect of intensity of light on stopping potential

3	Hall effect and Hall coefficient
4	No of lines/cm of grating using LASER
5	Solar cell characteristics
6	To Measure Numerical aperture and attenuation constant of optical fibre
7	Determination of TS, TDS and TSS in water sample
8	Determination of alkalinity in water
9	Determination of hardness of water
10	Determination of chloride in water
11	Determination of pH in water sample
12	Determination of biological oxygen demand

Text Books	
1	A Textbook of Engineering Physics, M N Avadhanulu, S Chand & Company Pvt. Ltd., (2014), ISBN: 978-8-121-90817-7
2	R.K.Gaurand S.L.Gupta, Engineering Physics. Dhanpat Rai and Sons Publications, (2012).
3	Engineering Chemistry: Fundamentals and Applications, Shikha Agarwal, Cambridge University Press, (2016) ISBN: 978-1-107-47641-7
4	Dr. S.S.Daraand Dr. S.S.Umare, "A Text book of Engineering Chemistry,"12th ed. S. Chand Publication, (2010). ISBN:978-81-219-0359-9
Reference Books	
1	Lasers and Non Linear Optics, B B Loud, New age international, (2011). ISBN:978-8-122-40324-4
2	Concepts of Modern Physics, Aurthur Beiser, McGrawhill, 6th Edition, (2009), ISBN:978-0-071-00144-1
3	Engineering Chemistry, K. Sesha Maheswaramma, Mridula Chugh, Pearson Education India, (2015), ISBN:978-8-131-77451-9
4	Engineering Chemistry, Satyaprakash & Manisha Agrawal, Khanna Book Publishing, (2012), ISBN-13. 978-9380016757
Web References	
1	Virtual LAB: https://www.vlab.co.in/participating-institute-amrita-vishwa-vidyapeetham
2	Physics Simulation: https://phet.colorado.edu/sims/html/plinko-probability/latest/plinkoprobability_en.html
3	Virtual LAB: https://vlab.amrita.edu/index.php?sub=1&brch=189&sim=343&cnt=1
4	Virtual LAB: https://ee1-nitk.vlabs.ac.in/List%20of%20experiments.html

Online Resources	
1	Virtual LAB: https://virtuallabs.merlot.org/vl_physics.html
2	Modelling & Simulation: https://cloud.scilab.in/

24UCSVS102	Android Programming	0/0/4
Nature of Course	Programming	
Prerequisites	Problem solving and Programming skills	
Course Objectives		
1	Understand Android fundamentals, setup, and basic app development.	
2	Design user-friendly interfaces and manage user interactions efficiently.	
3	Explore advanced topics like data handling, RESTful APIs, and scalability.	
4	Learn essential skills in notifications, location services, and Material Design.	
5	Gain hands-on experience in testing, debugging, and app deployment.	
Course Outcomes		
1	Demonstrate knowledge of Android development fundamentals, including setup, environment, and basic application development.	[U]
2	Apply critical thinking skills to design and develop user interfaces that enhance user experience and engagement.	[AP]
3	Analyze and implement advanced concepts such as data management, RESTful API integration, and scalability in Android applications.	[AP]
4	Evaluate and implement best practices in implementing notifications, location services, and adhering to Material Design guidelines.	[AP]
5	Synthesize learning outcomes by independently testing, debugging, and deploying Android applications on the Google Play Store, ensuring functionality and usability.	[AP]

Laboratory Component	
S.No	**List of Experiments**
1.	Install Android Studio, create a new project, and run a basic "Hello World" application on an emulator or a physical device.
2.	Design a simple layout with various UI components like buttons, text fields, and images. Implement onClickListeners to handle user interactions.

3.	Create an application with multiple fragments. Implement lifecycle methods like onCreateView(), onStart(), and onDestroy() to understand fragment lifecycle.
4.	Store user preferences using SharedPreferences. Build an app where users can set and retrieve their preferences.
5.	Create a database schema and perform CRUD (Create, Read, Update, Delete) operations using SQLiteOpenHelper and SQLiteDatabase.
6.	Fetch data from a public API (e.g., weather data or news feed) using Retrofit library. Parse JSON responses and display the data in the app.
7.	Develop an app to download an image from a URL using AsyncTask. Display the downloaded image in an ImageView.
8.	Integrate Google Maps API to display the user's current location on a map. Add markers and implement location updates.
9.	Create an app to send notifications at specified times using NotificationManager. Customize notification content and handle user interactions.
10.	Redesign an existing app using Material Design principles. Update UI components, colors, and typography to adhere to Material Design guidelines.
11.	Add animations and transitions to UI elements in an app. Experiment with different types of animations like fade, slide, and scale.
12.	Write unit tests for critical functions in your app using JUnit. Test various scenarios and ensure the correctness of the app's behavior.
13.	Use Espresso framework to write UI tests for your app. Test user interactions and verify UI components' states and behavior.
14.	Debug a sample Android app to identify and fix common issues like NullPointerException, IllegalStateException, and memory leaks.

Text Books	
1.	Reto Meier, "Android App Development for Dummies", 4th Edition, Wiley, (2019).
. 2.	Mark Murphy, "The Busy Coder's Guide to Android Development", CommonsWare, LLC, (2021).
3.	Bill Phillips, "Android Programming: The Big Nerd Ranch Guide", 4th Edition, Big Nerd Ranch, (2019).
Reference Books	
1.	Douglas Schmidt, "Android Programming: Concepts and Practice", Addison-Wesley Professional, (2013).
Web References	
1.	Android Developer Documentation: https://developer.android.com/develop
2.	https://onlinecourses.swayam2.ac.in/nou24_ge25/preview
Online Resources	
1.	https://www.coursera.org/learn/introduction-to-android-mobile-application-development

24UCSES103	Operating Systems	3/0/2
Nature of Course	Theory and Programming	
Prerequisites	Problem solving and Programming skills	

Course Objectives

1	To identify the structure and functions of Operating System.
2	To experiment CPU scheduling policies, synchronization techniques and deadlock handling in real time problems. To describe the OS mechanisms to handle processes and threads.
3	To articulate Memory management schemes.
4	To study various operating system concurrency and security issues.
5	To discuss Device Management, I/O and File systems concepts.

Course Outcomes

Upon completion of the course, students shall have ability to

1	Review the basic concepts and operations of operating systems.	[U]
2	Describe the Process and thread concepts in operating systems for real world problems.	[U]
3	Compare CPU scheduling algorithms, process synchronization mechanisms and deadlock handling methods.	[U]
4	Compare page replacement algorithms and do performance analysis.	[U]
5	Illustrate the concepts related to mass storage, I/O and file system.	[A]

Course Contents

Unit I Introduction to Operating System and Process Management (Process virtualization) 9 Hrs

Introduction to operating systems, CPU and Memory virtualization, Design goals of operating system, Process abstraction, Process creation, Process API, Process states, System calls for process management (fork (), wait (), exec ()), Limited direct execution (Process switching, Restricted Operations, Concurrency Issues), Necessity and Usage of threads, Thread creation, Uncontrolled Scheduling, Inter-process communication.

Case study

Linux Inter Process Communication, Android Inter Process Communication.

Unit II Process Synchronization 9 Hrs

Process execution mechanisms, Scheduling policies, Scheduling Metrics, FIFO, SJF, STCF, Priority, Round Robin, Multi-Level Feedback Queue (MLFQ), Changing and boosting priority, Issues with MLFQ, Scheduling proportional share and ticketing mechanism, Multiprocessor scheduling (Synchronization, Cache affinity, Single key scheduling, Multi queue scheduling), Challenges related to the Scheduling policies, Process Synchronization, Deadlocks, Deadlock in Multithreaded Applications, Deadlock Prevention, Deadlock Avoidance, Deadlock Detection, Deadlock Recovery.

Case study

Deadlock Detection in Windows10 operating system.

Unit III Main Memory Management 9 Hrs

Main memory, Contiguous Memory Allocation, Dynamic Memory Allocation, Paging, Structure of the Page Table Swapping, Segmentation, Memory allocation and free space management algorithms, Virtual memory, Demand Paging, Copy-On-Write, Page Replacement Allocation of Frames, Thrashing, VMS virtual memory system.

Case study

Acquiring live main memory contents in Windows 10.

Unit IV Concurrency and OS Security 9 Hrs

Introduction to Concurrency, Locks, Lock based concurrent data structures, Condition variables, Semaphores, Common concurrency problems, Event based concurrency, Operating system security, Data integrity and protection, Availability, Authentication, Botnets and worms, A Problem: Blocking System Calls, A Solution: Asynchronous I/O.

Case Study

Botnet Attacks against Linux Systems.

Unit V File Management, I/O and storage **9 Hrs**

File-System Interface, Creating, reading, and writing Files, Shared File Table Entries (Fork () and dup ()), Removing File, Directory operation (Creation, Deletion, Read), File organization-inode, multilevel index, File System Structure- Directory implementation- Allocation Methods- Free Space Management. File system Internals: File Systems - File System mounting - Partitions and Mounting. Disk free page management, File Sharing. I/O Systems: Overview - I/O Hardware. Mass Storage Structure: Overview - HDD Scheduling, HDD vs SSD, NVM Scheduling - Storage Device Management - Storage Attachment.

Case Study

Sun's network file system (NFS), Andrew File System (AFS), Interplanetary File System (IPFS)

Total Hours:	**45**

Text Books:	
1.	Remzi H. Arpaci-Dusseau and Andrea C. Arpaci-Dusseau, " Operating Systems: Three Easy Pieces", Arpaci-Dusseau Books, November, (2023) (Version 1.10).
2.	Abraham Silberschatz, Peter B. Galvin, Greg Gagne, "Operating System Concepts" 10th Edition, John Wiley, (2018).
3.	D.M Dhamdhere, "Operating Systems"- A Concept based Approach, 3rd Edition, McGraw Hill, (2017).
Reference Books:	
1.	Andrew S. Tanenbaum, Modern Operating Systems 5th Edition, Pearson Education, (2016).
2.	William Stallings, "Operating Systems – Internals and Design Principles", 8th Edition, Pearson Publications, (2014).
Web References:	
1.	https://www.studocu.com/sg/course/nanyang-technological-university/operating-systems/1390534
2.	https://www.cs.uic.edu/~jbell/CourseNotes/OperatingSystems/
3.	https://www.gatevidyalay.com/operating-system/
Online Resources:	
1.	https://www.coursera.org/learn/os-power-user
2.	https://nptel.ac.in/courses/106108101

Laboratory Component

S.No.	List of Experiments
1.	Identify the hardware configuration of your working machine using the /proc filesystem. a. Run command more /proc/cpuinfo and explain the following terms: processor and cores. Use the command lscpu to verify your definitions. You may want to understand the concept of CPU hyperthreading at a high level before attempting this question. b. How many cores does your machine have? c. How many processors does your machine have? d. What is the frequency of each processor? e. What is the architecture of your CPU? f. How much physical memory does your system have? g. How much of this memory is free? h. What is total number of number of forks and context switches since the system booted up?

2.	a. Write a program where a process forks a child. The child runs for a long time, say, by calling the sleep function. The parent must use the kill system call to terminate the child process, reap it, print a message, and exit. Understand how signals work before you write the program. b. Write a program that runs an infinite while loop. The program should not terminate when it receives SIGINT (Ctrl+C) signal, but instead, it must print "I will run forever" and continue its execution. You must write a signal handler that handles SIGINT in order to achieve this. So, how do you end this program? When you are done playing with this program, you may need to terminate it using the SIGKILL signal.
3.	c. Debug the pointers.cpp program given to you using GDB. The program contains some pointer related operations. A bug has been deliberately introduced in the code so that it generates a segmentation fault. Your task is to use GDB to find the line number of the wrong statement. Please make use of the GDB commands provided above in order to find the bug. d. Debug the fibonacci.cpp program given to you using GDB. This program is supposed to print fibonacci numbers until a certain value of n. However, there is a logical error introduced in the code which causes it to print wrong output. Your task is to use GDB to debug the program. You must insert suitable breakpoints, pause program execution, print intermediate values of variables from GDB, and monitor the execution step by step in order to find the logical error. Even if you can identify the error without stepping through the code, you must be able to demonstrate the process of debugging using GDB during your evaluation.
4.	a. Write a simple C/C++ program that runs for a long duration, say, by pausing for user input or by sleeping. While the process is active, use the ps or any other similar command with suitable options, to measure the memory usage of the process. Specifically, measure the virtual memory size (VSZ) of the process, and the resident set size (RSS) of the process (which includes only the physical RAM pages allocated to the process). You should also be able to see the various pieces of the memory image of the process in the Linux proc file system, by accessing a suitable file in the proc file system. b. Now, add code to your simple program to memory map an empty page from the OS. For this program (and this lab, in general), it makes sense to ask the OS for an anonymous page (since it is not backed by any file on disk) and in private mode (since you are not sharing this page with other processes). Do not do anything else with the memory mapped page. Now, pause your program again and measure the virtual and physical memory consumed by your process.
5.	Five philosophers dine together at the same table. Each philosopher has his own plate at the table. There is a fork between each plate. The dish served is a kind of spaghetti which has to be eaten with two forks. Each philosopher can only alternately think and eat. Moreover, a philosopher can only eat his spaghetti when he has both a left and right fork. Thus two forks will only be available when his two nearest neighbors are thinking, not eating. After an individual philosopher finishes eating, he will put down both forks. Implement a program (a concurrent algorithm) such that any philosopher will not starve; i.e., each can forever continue to alternate between eating and thinking, assuming that no philosopher can know when others may want to eat or think.
6.	Implementation of Banker's Algorithm for Deadlock Avoidance.
7.	My friend John is an "environmental" activist and a humanitarian. He hated the ideology of Thanos from the Avengers: Infinity War. He sucks at programming. He used too many variables while writing any program. One day, John gave me a memory dump and asked me to find out what he was doing while he took the dump. Use volatility tool and analyse the given memory dump for the same.

8.	Implementation of Page Replacement Techniques.
9.	a. Write a program that has a counter as a global variable. Spawn 10 threads in the program, and let each thread increment the counter 1000 times in a loop. Print the final value of the counter after all the threads finish—the expected value of the counter is 10000. Run this program first without using locking across threads, and observe the incorrect updation of the counter due to race conditions (the final value will be slightly less than 10000). Next, use locks when accessing the shared counter and verify that the counter is now updated correctly. b. Write a program where the main default thread spawns N threads. Thread i should print the message "I am thread i" to screen and exit. The main thread should wait for all N threads to finish, then print the message "I am the main thread", and exit.
10.	Implementation of Disk Scheduling Algorithms. Compare CPU and Disk Scheduling algorithms in terms of the number of voluntary and involuntary context switches.

24UCSPC101	Digital Logic and Computer Architecture	3/0/0
Nature of Course:	Theory	
Prerequisites:	Foundation of system and design.	

Course Objectives

1	To make students know about the Parallelism concepts in Programming.
2	To give the students an elaborate idea about the different memory systems and buses.
3	To introduce the advanced processor architectures to the students.
4	To make the students know about the importance of multiprocessor and multicomputer.
5	To study about data flow computer architectures

Course Outcomes

1	Understand the concepts of parallelism in hardware/software.	[U]
2	Analyse memory organization and mapping techniques.	[A]
3	Describe architectural features of advanced processors	[A]
4	Design combinational and sequential circuits and using FSM methods.	[A]
5	Simulate digital circuits using Hardware descriptive language.	[A]

Course Content

Unit I Digital System and Binary Numbers **9 Hrs**

Introduction to gates, Review of Boolean algebra and De-Morgan's law, error detection and error correction codes. Boolean function operations, digital logicgates.

Case study

Design circuit seven segment display using VHDL

Unit II Combinational Logic Circuits and Gate Level Minimization **9 Hrs**

Design procedure of combinational circuits, SOP, Canonical SOP, the k-map method, NAND and NOR implementation Ripple carry adder and Carry look ahead adder, Binary number systems, Signed binary numbers, Binary arithmetic, binary codes, Binary Adder, Binary Subtractor, Binary Multiplier.

Case study

Design of Combinational Circuits using Programmable Logic Devices (PLDs): Programmable Read Only Memories (PROMs)

Unit III Sequential Logic Circuits and Computer Architecture **9 Hrs**

Latches: RS latch and JK latch, Flip-flops-RS, JK, T and D flip flops, Master-slave flip flops, Edge-triggered flip-flops. Analysis and Design of Synchronous Sequential Circuits: Introduction to sequential circuits, Characteristic table, Characteristic equation and Excitation table. Computer Architecture, Flynn’s Classification of Computers, Metrics for Performance Measurement. Advanced Optimizations of Cache Performance, Memory Technology and Optimizations, Virtual Memory and Virtual Machines, The Design of Memory Hierarchy, Introduction to Pin Instrumentation and Cache grind,

Case Study

Memory Hierarchies in Intel Core i7 and ARM Cortex-A8, Traffic Light Controller with Sequential logic.

Unit IV Thread Level Parallelism **9 Hrs**

Introduction, RISC and CISC, Shared-Memory Multicore Systems, Performance Metrics for Shared-Memory Multicore Systems, Cache Coherence Protocols, Synchronization, Memory Consistency, Multithreaded Programming using OpenMP,

Case Study

Intel Skylake and IBM Power8.

Unit V Data Level Parallelism **9 Hrs** Introduction, Vector Architecture, SIMD Instruction Set Extensions for Multimedia, Graphics Processing Units, GPU Memory Hierarchy, Detecting and Enhancing Loop- Level Parallelism, CUDA Programming, **Case Study** Nvidia Maxwell.	
Total Hours:	**45**

24UCSPC102	Python Programming	3/0/2
Nature of Course	Theory and Programming	
Prerequisites	NIL	

Course Objectives

1	To learn various operators and control statements of Python Programming.
2	To understand Python data structures and string manipulation techniques.
3	To gain knowledge of file handling and manage exceptions effectively in Python.
4	To learn the principles of object-oriented programming (OOP) in Python.
5	To acquire advanced Python programming techniques and libraries for data analysis and visualization.

Course Outcomes
Upon completion of the course, students shall have ability to

1	Recall the fundamental concepts and be able to list, define, and identify basic Python syntax and data types for given problems.	[U]
2	Demonstrate the data structures and string manipulation techniques proficiently in Python programming tasks.	[A]
3	Apply file handling operations and exception handling strategies in Python.	[A]
4	Discover object-oriented solutions in Python, including class hierarchies, inheritance, polymorphism, and encapsulation.	[A]
5	Develop practical applications, such as web development, data analysis, automation scripts, and scientific computing projects.	[A]

Unit I Introduction to Python **9 Hrs**
Introduction to Python and its features, setting up the Python environment, basic syntax, variables, and data types, operators (arithmetic, relational, logical, bitwise), input and output operations, control flow: if, elif, else statements, loops: for, while loops, functions: defining and calling functions, return statement, scope of variables.
Case Study
Develop a calculator in Python with error handling capability (for division by zero and invalid inputs).

Unit II Data Structures and Strings **9 Hrs**
Lists: creation, indexing, slicing, methods, Tuples: creation, indexing, methods, Dictionaries: creation, accessing elements, methods, Sets: creation, methods, set operations, Strings: string operations, methods, string formatting, list comprehensions and generator expressions.
Case Study
Develop a Student Record Management System using Python.

Unit III File and Exception Handling **9Hrs**
File operations: opening, reading, writing, and closing files, different modes of file operations (read, write, append, etc.), handling exceptions using try, except, finally blocks, custom exceptions.
Case Study
Develop a Library Management System using Python.
Unit IV Object-Oriented Programming (OOP) **9 Hrs**
Introduction to OOP concepts: classes, objects, inheritance, polymorphism, encapsulation, defining classes and creating objects, methods and constructors, inheritance: single and multiple inheritance, method overriding and overloading, polymorphism, encapsulation and data hiding.
Case Study
Create a banking system using OOP concepts that supports operations like creating accounts, depositing money, withdrawing money, and checking balance. Design classes for different types of accounts (savings, current) using inheritance to encapsulate account data securely.

Unit V Advanced Python Libraries **9 Hrs**

Introduction to modules and packages, using standard libraries (os, sys, math, datetime), introduction to third-party libraries (NumPy, pandas, matplotlib), working with dates and times, basic concepts of web scraping using BeautifulSoup, introduction to GUI programming with Tkinter.

Case Study

PBL: Collect and analyze a dataset using pandas and NumPy for data manipulation and numerical operations, then visualize the data with matplotlib and create a simple GUI using Tkinter to display the results.

Total Hours (Theory)	**45**

Lab Component

S.No.	Lab Exercise
1.	Set up the Python environment and execute a basic Python script.
2.	Program to explore different data types and perform operations on variables.
3.	Program to implement if-elif-else statements and loops (for, while) to solve simple problems.
4.	Program to create and use functions, including defining parameters, return statements, and understanding variable scope.
5.	Program to perform indexing, slicing, and use methods on lists and tuples.
6.	Program to manipulate dictionaries and sets using their methods and operations.
7.	Program to utilize string methods, string formatting, and manage text data efficiently.
8.	Program to implement list comprehensions and generator expressions for efficient data processing.
9.	Program to read from and write to files in various modes (read, write, append).
10.	Program to use try-except blocks to handle exceptions and create custom exceptions.
11.	Program to define classes, create objects, and use constructors and methods, demonstrating encapsulation.
12.	Program to implement single and multiple inheritance, demonstrating method overriding and polymorphism.
13.	Program to create and use modules and packages to organize code efficiently and use standard libraries such as os, sys, math, and datetime.
14.	Program to analyze datasets using NumPy for numerical operations and pandas for data manipulation, and create various plots to visualize data using matplotlib.
15.	Program to scrape and parse data from a website using BeautifulSoup.
	Total Hours (Lab) **30**
	Total Hours (45+30) **75**

Text Books

1.	Wesley J. Chun, “Core Python Applications Programming”, 3rd Edition , Pearson Education, (2016).
2.	Eric Matthes, “Python Crash Course”, 2nd Edition, No Starch Press, (2019).
3.	Lambert, Fundamentals of Python: First Programs with MindTap, 2nd edition , Cengage Learning publication.
4.	Charles Dierbach, “Introduction to Computer Science using Python”, Wiley, (2015).
5.	Charles Severance, “Python for Everybody”, 1st Edition, CreateSpace Independent Publishing Platform, (2016).

Reference Books

1.	Al Sweigart, “Automate the Boring Stuff with Python”, 2nd Edition, No Starch Press, 2019.
2.	Downey, A. et al., “How to think like a Computer Scientist: Learning with Python”, John Wiley, (2015).

Web References

1.	Official Python Documentation: https://www.python.org/
2.	MIT OpenCourseWare - Introduction to Computer Science and Programming in Python: https://ocw.mit.edu/courses/6-0001-introduction-to-computer-science-and-programming-in-python-fall-2016/

Online Resources

1.	W3Schools Python Tutorial: https://www.w3schools.com/python/
2.	Coursera - Python Data Structures: https://www.coursera.org/learn/python-data

3.	edX - Introduction to Python: Absolute Beginner: https://www.edx.org/learn/python
4.	Stanford University - CS106A: Programming Methodology: https://cs106a.stanford.edu/

SEMESTER - III

24UCSOE201	Renewable Energy Systems	3/0/0
Nature of Course	Theory	
Prerequisites	Applied Science	
Course Objectives:		
1	To know the present status of Indian and global energy scenario.	
2	To learn the various solar energy technologies and its applications.	
3	To educate the various wind energy technologies.	
4	To explore the various bio-energy technologies.	
5	To study the ocean and geothermal technologies.	
Course Outcomes (Theory) **Upon completion of the course, students shall have ability to**		
1	Illustrate the Indian and global energy scenario	[U]
2	Compare various solar energy technologies and identify its applications.	[A]
3	Infer wind data and compare various wind energy systems.	[A]
4	Examine various bio-energy technologies and identify their application.	[A]
5	Interpret ocean and geothermal energy conversion technologies.	[A]
Course Contents		

UNIT I ENERGY SCENARIO **9 Hrs**

Indian energy scenario in various sectors,domestic, industrial, commercial, agriculture, transportation and others ,Present conventional energy status ,Present renewable energy status,Potential of various renewable energy sources,Global energy status,Per capita energy consumption ,Future energy plans

UNIT II SOLAR ENERGY **9 Hrs**

Solar radiation , Measurements of solar radiation and sunshine, Solar spectrum ,Solar thermal collectors,Flat plate and concentrating collectors, Solar thermal applications , Solar thermal energy storage , Fundamentals of solar photo voltaic conversion ,Solar cells ,Solar PV Systems,Solar PV applications

UNIT III WIND ENERGY **9 Hrs**

Wind data and energy estimation – Betz limit – Site selection for wind farms – characteristics – Wind resource assessment – Horizontal axis wind turbine – components – Vertical axis wind turbine – Wind turbine generators and its performance – Hybrid systems–Environmental issues Applications

UNIT IV BIO-ENERGY **9 Hrs**

Bio resources , Biomass direct combustion , thermochemical conversion, biochemical conversion-mechanical conversion ,Biomass gasifier ,Types of biomass gasifiers ,Cogeneration ,Carbonisation , Pyrolysis ,Biogas plants ,Digesters ,Biodiesel production, Ethanol production ,Applications.

UNIT V OCEAN AND GEOTHERMAL ENERGY **9 Hrs**

Small hydro ,Tidal energy,Wave energy,Open and closed OTEC Cycles,Limitations ,Geothermal energy,Geothermal energy sources ,Types of geothermal power plants ,Applications Environmental impact.

	Total Hours	**45**
Text Books		
1	Godfrey Boyle, “Renewable Energy, Power for a Sustainable Future”, Oxford University Press,U.K., (2012).	
2	David M. Buchla., “Renewable Energy Systems”, Pearson education publication, Hard cover/Paperback (2017).	
3	Sukhatme.S.P., “Solar Energy: Principles of Thermal Collection and Storage”,Tata McGraw Hill Publishing Company Ltd., New Delhi,(2009).	
Reference Books		
1	TiwariG.N.,“Solar Energy–Fundamentals Design, Modelling and applications”, Alpha Science Intl Ltd, (2015).	
2	Mehmet Kanoglu “Fundamentals and Applications of Renewable Energy”, Indian edition McGraw Hill Publication, Hard cover/Paperback (2020).	
3	Twidell,J.W.&WeirA.,“Renewable Energy Resources”, EFN Spon Ltd., UK, (2015).	
Web References		
1	https://onlinecourses.nptel.ac.in/noc22_ch27/preview	
2	https://www.coursera.org/courses?query=renewable%20energy	
3	https://www.slideshare.net/Akshayro/renewable-energy-79579205	
Online Resources		
1	https://www.youtube.com/watch?v=VfowJHJz6-s	
2	https://www.youtube.com/watch?v=44Wp3WE1AHs	
3	https://www.youtube.com/watch?v=uStFvcz9Or4	

24UCSMD201	**Financial Management**	**2/0/0**

Nature of Course:	Theory	
Prerequisites:	NIL	
Course Objectives:		
1	To develop financial literacy among engineering minds.	
2	To enhance financial decision-making skills among young minds	
3	To create financial planning and budgeting methods in day-to-day routine	
4	To Understand Investment methodology and risk management skills	
5	To promote ethical financial practices	
Course Outcomes		
Upon completion of the course, students shall have ability to		
1	Understand the importance of financial management from a strategic	[U]
2	Understand the financial statement and the component impacting financial	[U]
3	Apply the budgeting techniques in financial decisions making	[AP]
4	Analyze the short-term and long-term sources of finance	[A]
5	Analyze the decision of capital structure and distribution of dividend	[A]

Course Contents

Unit I Introduction to Financial Statements **6 Hrs**

Introduction to Finance, various types of business, role of finance in business, importance and financial decision making, basic finance, revenue, expenses, profit, and loss Assets, liabilities, and equity, time value of money.

Case Study

Creation of balance sheet for business and exercises on calculating present and future value.

Unit II Understanding Financial Statements **6 Hrs**

Introduction to Financial Statement, Income Statement, Revenue and expenses ,Gross profit, operating profit, and net profit, Balance Sheet, Assets: current and noncurrent,Liabilities: current and long-term ,Equity: shareholder's equity and retained earnings, Cash Flow Statement , Operating activities, Investing activities, Financing activities

Case Study

Hands-on practice with sample financial statements, analyze a company's financial health and Preparing a brief report on the financial health of the company.

Unit III Financial Planning and Budgeting **6 Hrs**

Introduction to Budgeting, purpose and benefits of budgeting, different types of budgets, steps to create a budget -using historical data and forecasting, financial planning, short-term vs. long-term planning, forecasting financial performance, setting financial goals.

Case Study

Creating a business budget, develop a budget for a hypothetical project and create a financial plan for a small business

Unit IV Investment Principles **6 Hrs**

Introduction to Investments, types of investments, stocks, bonds, mutual funds, real estate, Financial Markets, types process, benefits in economic growth, Ethics in Finance, importance of ethical finance, common ethical issues and dilemmas, Risk and Return, Understanding risk, Measuring return-risk-return tradeoff.

Case Study

Simulation game on building a diversified portfolio and group activity on assessing investment risk.

Unit V Corporate Finance **6 Hrs**

Introduction to corporate finance, role of corporate finance in business, financial management objectives, sources of corporate finance, debt financing, equity financing, capital structure, optimal capital structure, trade-off between debt and equity-dividend policy-types of dividends factors influencing dividend policy.

Case Study

Analyzing a company's capital structure and dividend policy and Preparing a report on financing options. for a company.

Total Hours:	**30**

Text Books:	
1.	I M. Pandey Financial Management, Vikas Publishing House Pvt. Ltd., 11th edition, (2018).
2.	Financial Management: Principles and Applications" by Sheridan Titman, Arthur J. Keown, and John D. Martin, Pearson, 13th Edition, (2019).
3.	Fundamentals of Corporate Finance" by Stephen A. Ross, Randolph W. Westerfield, and Bradford D. Jordan, McGraw-Hill Irwin, 13th Edition, (2022).
4.	Corporate Finance: Theory and Practice" by Aswath Damodaran, Wiley Series in Finance India, 2nd Edition, (2007).
Reference Books:	
1.	M.Y. Khan and P.K.Jain Financial management, Text, Problems and cases Tata McGraw Hill, 8th edition, (2017).
2.	Financial Statement Analysis and Security Valuation" by Stephen H. Penman , McGraw-Hill Irwin, 4th Edition, (2017).
3.	Budgeting Basics and Beyond" by Jae K. Shim and Joel G. Siegel, Wiley, 4th Edition , (2016).

Web References:	
1.	https://www.coursera.org/specializations/financial-management
2.	https://www.udemy.com/course/the-complete-finance-managercourse/ course/?couponCode=NVDPRODIN35
3.	https://onlinecourses.nptel.ac.in/noc21_mg06/preview
4.	https://alison.com/tag/financial-management#google_vignette
5.	https://www.mygreatlearning.com/academy/learn-for-free/courses/introduction-to-financial-management
Online Resources:	
1.	https://www.investopedia.com/best-resources-for-improving-financial-literacy-5091689
2.	https://www.dividend.com/my-money/50-free-resources-to-help-manage-your-money/
3.	https://fastercapital.com/content/The-best-online-resources-for-finance.html
4.	https://researchguides.austincc.edu/c.php?g=434734&p=5826974
5.	https://www.collegeconsensus.com/careers/free-resources-for-financial-managers/

24UCSAE201	Analytical Skill Development	2/0/0
Nature of Course	Theory	
Prerequisites	NIL	
Course Objectives		
1	To develop clarity in the exploration process of student career and to match his skills and interests with a chosen career path.	
2	To develop required aptitude skills for carrier enhancement	
3	To design the functional and chronological resume	
4	To demonstrate the importance of critical thinking ability and expression in group discussions	
5	To prepare students for the various professional interviews	
Course Outcomes (Theory) **Upon completion of the course, students shall have ability to**		
1	Understand to the problem-solving skill for carrier enhancement.	[U]
2	Understand the students able to prepare resume for personal skill develop	[U]
3	Apply the knowledge of GD & Presentation Skill during Industry Assessments for Placement/Internship/Industry Training/Higher Studies/Competitive Exams etc.	[AP]
4	Analyze and apply the critical thinking ability as required during Aptitude/Technical Tests.	[A]
5	Develop the logical and programming solving skill of students.	[A]

Course Contents

UNIT I General Aptitude **6 Hrs**

Mixture & Allegation, Bankers Discount, Chain Rule, Percentage, Problems on Ages, Profit & Loss, Simple Interest, Time & Distance

Case study

Students needed to write 2 Aptitude Test based unit 1

UNIT II Numerical Aptitude **6 Hrs**

Area, Boats and Streams, Clocks, LCM and HCF, Permutation & Combination, Time & Work, Profit & Loss

Case Study

Students needed to write 2 Aptitude Test based unit 2

UNIT III Group Discussion **6 Hrs**

Why GDs are implemented commonly, Aspects which make up a Group Discussion, Tips on group discussion, do's and don'ts of GD and Presentation skills.

Case Study

Team based two group discussion must be conducted based on latest topics.

UNIT IV Logical Reasoning **6 Hrs**

Coding and Decoding (Visual Reasoning and series), Statement & Conclusions (Syllogisms), Relationships (Analogy), Data arrangements, Crypt arithmetic. Data Interpretation & Data Sufficiency, Blood relation and dices, Clocks and Calendar, Direction sense and cubes, Logical connectives.

Case Study

Students needed to write 2 test based on logical reasoning topics. **UNIT V Coding and Problem Solving** **6 Hrs** CodeChef, HackerRank, LeetCode, LLC, Coderbyte, Topcoder **Case Study** Students needed to participate online coding event and increase rating.	
Total Hours	**30**
Text Books	
1	A Modern Approach to Verbal & Non-Verbal Reasoning by R.S. Agarwal.
2	Reasoning verbal and Non-Verbal by B. S. Sijwali.
3	Master the Group Discussion & Personal Interview - Complete Discussion on the topics asked by reputed B-schools & IIMs by Sheetal Desarda.
Web References	
1	https://leverageedu.com/blog/general-aptitude/
2	https://prepinsta.com/important-aptitude-topics-for-campus-placements/
3	https://www.toppr.com/guides/quantitative-aptitude/
Reference Books	
1	Shortcuts in Reasoning (Verbal, Non-Verbal, Analytical)(2018)
2	Analytical Reasoning by M. K. Panday(2009).
3	Logical and analytical reasoning by K. Gupta(2020).
4	Multi-dimensional reasoning by Mishra & Kumar Dr. Lal(2010).
Online Resources	
1	https://www.practiceaptitudetests.com/non-verbal-reasoning-tests/
2	https://www.educationquizzes.com/11-plus/non-verbal-reasoning/
3	https://www.livecareer.com/resume/examples/web-development/e-learning-developer
4	https://novoresume.com/career-blog/how-to-write-a-resume-guide
5	https://www.coursera.org/learn/resume-writing

24UCSVE201	Foreign Language - German- I	0/0/4
Nature of Course	Practical	
Prerequisites	NIL	

Course Objectives:

1	Learn introductory greetings and farewells in German.
2	Formulate and respond to basic questions about oneself and others in German.
3	Discuss hobbies and interests using essential German vocabulary.
4	Navigate a German restaurant by ordering food and requesting the bill.
5	Comprehend and utilize key German phrases for everyday situations.

Course Outcomes:
Upon completion of the course, students shall have ability to

1	Interpreting common German greetings and farewells.	[U]
2	Comprehend and explain the structure of basic German questions for self and others	[U]
3	Choosing German vocabulary to discuss personal hobbies and interests in simple conversations.	[AP]
4	Distinguish between different food and beverage options on a German menu.	[A]
5	Illustrate the appropriateness of different German phrases for various everyday scenarios.	[A]

Laboratory Component

S.No.	Lab Exercise
1	Greet, say goodbye, and inquire about well-being in German.
2	Introduce yourself, state origin, and ask/receive someone's name formally.
3	Spell your name and provide your phone number in German.
4	Discuss hobbies with someone using basic German vocabulary.
5	Navigate a German restaurant by ordering coffee and using essential phrases.
6	Select and order a meal from a provided German menu.
7	Request the bill at a restaurant using a formal German expression.
8	Express gratitude for the meal using a German phrase.
9	Ask for directions to a specific location using appropriate German phrases.
10	Inquire about someone's destination using a German question.
11	Inform someone of your destination using a set German phrase.
12	Ask for the time using a standard German expression.
13	Request clarification on a German term or phrase using a formal expression.
14	Count from 1 to 10 in German.
15	State one's birthday using German numerals and date format.

Text Books:

1.	Deutsch im Blick - Strong foundation in grammar and vocabulary for reading comprehension.

2.	Complete German - Structured approach with interactive exercises.
3.	Menschen A1.1 - Communicative approach for real-life situations.
Reference Books:	
1.	Langenscheidt Dictionary - Comprehensive German-English dictionary.
2.	New German Grammar - Detailed explanations of grammar rules.
Web References:	
1.	Goethe-Institut Online Courses - Interactive exercises and grammar explanations.
2.	DW Learn German - Video lessons, podcasts, and vocabulary trainers.
Online Resources:	
1.	Memrise - Flashcards for memorizing vocabulary.
2.	Duolingo - Gamified app for basic phrases and grammar.

24UCSPR201	Seminar- I	0/0/4
Nature of Course	Theory	
Prerequisites	NIL	

Evaluation through Seminar Presentation at the Department

The student will give a seminar based on his training report, before an expert committee constituted by the concerned department as per norms of the institute. The evaluation will be based on the following criteria:

1. Depth of knowledge and skills
2. Communication & Presentation Skills
3. Team Work
4. Creativity
5. Planning & Organizational skills
6. Adaptability
7. Analytical Skills
8. Attitude & Behavior at work
6. Societal Understanding
7. Ethics
8. Regularity and punctuality
9. Attendance record
10. Diary/Work book
11. Student's Feedback from External Internship Supervisor

After completion of Internship, the student should prepare a comprehensive report to indicate what he has observed and learnt in the training period.

Internship Diary/workbook may be evaluated on the basis of the following criteria:

1. Proper and timely documented entries
2. Adequacy & quality of information recorded
3. Data recorded
4. Thought process and recording techniques used
5. Organization of the information

The report shall be presented covering following recommended fields but limited to,

1. Title/Cover Page
2. Internship completion certificate
3. Internship Place Details- Company background-organization and activities/Scope and object of the study / Supervisor details
4. Index/Table of Contents
5. Introduction
6. Title/Problem statement/objectives
7. Motivation/Scope and rationale of the study
8. Methodological details
9. Results / Analysis /inferences and conclusion
10. Suggestions / Recommendations for improvement to industry, if any
11. Attendance Record
12. Acknowledgement
13. List of reference (Library books, magazines and other sources)

Feedback from internship supervisor (External and Internal)

Post internship, faculty coordinator should collect feedback about student with recommended parameters include as- Technical knowledge, Discipline, Punctuality, Commitment, Willingness to do the work, Communication skill, individual work, Team work, Leadership.

24UCSPC206	Computer Networks	3/0/0
Nature of Course:	Theory	
Prerequisites:	Discrete Structure	

Course Objectives:

1	To study networking fundamentals
2	To learn functionality of physical and data link layer
3	To understand packet delivery system and various protocols associated with it
4	To study application layer functionality
5	To learn and understand computer network security

Course Outcomes

Upon completion of the course, students shall have ability to

1	Understand networking fundamental concepts	[U]
2	Apply error detection techniques for error free data delivery	[AP]
3	Compare various addressing techniques	[A]
4	Apply corrective measures to secure email delivery process	[AP]
5	Configure a firewall to restrict unwanted and hazardous traffic	[A]

Unit I Networking fundamentals **9 Hrs**

Network edge, Internet vs intranet, Network core, Network topology, Delay, Loss, and Throughput in Networks, Networking devices, ISO-OSI reference model, TCP-IP Model.

Case Study

Simulating a Local Area Network

Unit II Physical Layer and Data Link Layer **9 Hrs**

Data and Signals, Bandwidth Utilization: Multiplexing and Spreading, Transmission Media, Switching, Error Detection and Correction, Data Link Control, Multiple Access, Wired LANs, Wireless LANs Connecting LANs, Backbone Networks, and Virtual LANs.

Case Study

Simulating Wi-Fi networks

Unit III Network Layer and Transport Layer **9 Hrs**

Logical addressing, Internet protocol, Address Mapping, Error Reporting, Multicasting, Packet delivery, Forwarding, routing, Process-to-Process Delivery: UDP, TCP, and SCTP, Congestion Control and Quality of service.

Case Study

TCP simulation

Unit IV Application Layer **9 Hrs**

Domain Name, Remote Multimedia Logging, Electronic Mail, and File Transfer, WWW and HTTP, Network Management: SNMP, Multimedia, Voice over IP.

Case Study

Simulation of a Satellite Network.

Unit V Security in Computer Networks	**9 Hrs**
Message Integrity and Digital Signatures, End-Point Authentication, Securing E-Mail, Securing TCP Connections: SSL, Network-Layer Security: IPsec and Virtual Private Networks, Securing Wireless LANs, Operational Security: Firewalls and Intrusion Detection Systems **Case study** Securing an Application Layer Gateway: An Industrial perspective.	
Total Hours:	**45**

Text Books	
1.	Kurose K R. "Computer networking: A top-down approach by james", Kurose, Keith W. Ross, 7th edition, (2017).
2.	Forouzan, Behrouz A. Data communications and networking. Huga Media, (2007).
3.	Tanenbaum, Andrew S. Computer networks. Pearson Education India, (2003)
Reference Books	
1.	Kiser, Quinn. Computer networking and cybersecurity: a guide to understanding communications systems, internet connections, and network security along with protection from hacking and cyber security threats. 2020.
Web References	
1.	https://www.vssut.ac.in/lecture_notes/lecture1423905560.pdf
2.	https://ocw.mit.edu/courses/6-829-computer-networks-fall-2002/pages/lecture-notes/
Online Resources	
1.	https://nptel.ac.in/courses/106/104/106104148/
2.	https://nptel.ac.in/courses/106/104/106104028/

24UCSPC202	Data Structures and Algorithms	3/0/2
Nature of Course:	Theory & Programming	
Prerequisites:	Problem solving and Programming skills	
Course Objectives:		
1	To learn asymptotic notations and representation of structured data.	
2	To construct and perform various operations on Tree.	
3	To represent data as per the problem statement using Graph data structure.	
4	To learn various searching and sorting techniques.	
5	To represent, retrieve and search specific data using hash table .	
Course Outcomes		
1	**Understand** asymptotic notations and representation of structured data.	[U]
2	**Interpret** various operations on trees for different problem statement.	[A]
3	**Construct** solution for problem using Graph data structure.	[A]
4	**Compare** various searching and sorting algorithms.	[A]
5	**Illustrate** various hashing techniques to represent data in hash table.	[A]

Course Contents

Unit I Linear Data Structure **9 Hrs**

Introduction: Time and space complexity analysis, Abstract Data Type (ADT), the List ADT, Array Implementation, Linked List Implementation, The Stack ADT, The Queue ADT, Applications of Stack, Queue, and List.

Case Study

Efficient Data Storage and Retrieval - Develop an efficient data storage and retrieval system for a news feed that includes posts, comments, and likes.

Unit II Trees **9 Hrs**

Tree Traversal, Binary Trees, Definitions, Expression Tree, Binary Tree Traversals, The Search Tree ADT, Binary Search Trees, AVL Tree, Splay Tree, B-Tree, Red Black Tree, Self-Balancing Trees, Segment Trees.

Case Study

I) Huffman Encoding - Implement Huffman Encoding for data compression.

II) Tree Data Structure in a Social Network - Explore how trees are used in social networks to manage hierarchical relationships.

Unit III Graphs **9 Hrs**

Weighted and Directed graphs, Adjacency matrix and list implementation, Traversal – Breadth First Search & Depth First Search, Minimum Spanning Tree, Topological Sorting, Connectivity in Graph, Maximum Flow in Graph.

Case Study

World Wide Web, Facebook, Google map - Apply graph algorithms to optimize network based applications.

Unit IV Sorting and Searching Techniques **9 Hrs**

Insertion sort, Selection sort, Shell Sort, Bubble sort, Quick sort, Heap sort, Merge sort, Radix sort, Stable sorting algorithms, Comb sort, Radix Sort, Searching, Linear search, Binary search, Interpolation Search, Exponential Search.

Case Study

For sorting student grades - Implement various sorting algorithms to manage and sort student grades efficiently. Compare and analyze different sorting techniques for performance and stability.

Unit V Hashing Techniques	**9 Hrs**
Direct Address Table, Hash function, Collision resolution techniques, Linear Probing, Quadratic Probing, Double Hashing, Open Addressing. **Case Study** Hashing Technique in a Ticketing System - Design a robust ticketing system for events using hashing techniques.	
Total Hours:	**45**

Laboratory Component

S.No	List of Experiments
1.	A library needs to manage a collection of books (data fields such as book title, author, ISBN, availability status, and other relevant information), maintain information about each book, support operations like adding, removing, and searching for books, and provide efficient access to library resources. Use the Concept of Linked List, binary search and suitable sorting technique for implementation of Library Management system.?
2.	Design and implement a family tree hierarchy using a tree data structure. The family tree should allow efficient storage, retrieval, and manipulation of information related to individuals and their relationships within a family.
3.	Design and implement Employee Management system tree hierarchy using a binary search tree data structure. The family tree should allow efficient storage, retrieval, and manipulation of information related to individuals and their relationships within the employee. The insertion of new employee carried out based on their salary.
4.	There are flight paths between cities. If there is a flight between city A and city B then there is an edge between the cities. The cost of the edge can be the time that flight takes to reach city B from A or the amount of fuel used for the journey. The node can be represented by airport name or name of the city. Use adjacency list representation of the graph and adjacency matrix representation of the graph. Justify the storage representation used.
5.	You are a job seeker preparing for interviews in different locations. You have a list of cities where interviews are scheduled, and you want to find the shortest path to visit all the interview locations, starting from your current location. The objective is to minimize the total travel distance and time while ensuring you reach each interview location on time. Use appropriate data structure and algorithm to implement it.
6.	You are planning your wedding and need to select a marriage hall location from a list of available options. You want to find the shortest path from your current location to visit all the marriage hall locations and determine the optimal route that minimizes travel distance and time. Use appropriate data structure and algorithm to implement it.
7.	A Music Library System is a software application used to manage the storage and retrieval of music tracks and albums. Hash tables can be used as a data structure to efficiently store and retrieve music track and album information
8.	Medical Record System is a software application used to manage medical records, including patient information, medical history, and treatment plans. Hash tables can be used as a data structure to efficiently store and retrieve medical records.

Text Books

1.	Horowitz and Sahani, Fundamentals of Data Structures, Galgotia Publisher,
2.	Sartaj Sahani,Data Structures, Algorithms and Applications in C++, 2nd edition, Universities Press,ISBN-81-7371-522
3.	Holzner, Steven ,"Java 2 Programming: Black Book ", 5th edition Dreamtech

Reference Books

1.	Narasimha Karumanchi, " Data Structures And Algorithms Made Easy", N. K. (2017).

Web References

1.	https://in.coursera.org/learn/advanced-data-structures
2.	https://onlinecourses.nptel.ac.in/noc23_cs15/preview
Online Resources	
1.	https://www.coursera.org/specializations/data-structures-algorithms

24UCSPC203	Java Programming	3/0/2
Nature of Course	Theory & Programming	
Prerequisites	NIL	
Course Objectives		
1.	To understand common Java libraries, operators, and decision statements.	
2.	To Understand the concepts of control statements and their role in building complex ML systems.	
3.	To introduce Java string libraries for developing memory efficient applications.	
4.	To Utilize Java streams to develop concise and efficient in AI applications.	
5.	To introduce java event handler for interactive programming.	
Course Outcomes		
Upon completion of the course, students shall have ability to		
1	Apply Java Programming Fundamental Concepts For Application Development.	[AP]
2	Apply Java control statements for user friendly application development.	[AP]
3	Develop memory efficient application using Java string.	[AP]
4	Apply Java stream to make the code more concise and efficient.	[AP]
5	Apply Java event handler to controls the event and decides what should happen if an event occurs in the code.	[AP]

Course Contents

Unit I Introduction to Java Programming **9 Hrs**

Introduction to Java: Java Architecture- JVM, JRE & JDK, Keywords, Features of Java, Console input and output statements, variables and Identifiers, Scope of Variables, Data types, Type Conversion, Comments, Command Line Arguments, Access Modifiers Operators, Unary Operator, Arithmetic Operator, Shift Operator, Relational Operator, Bitwise Operator ,Logical Operator ,Ternary Operator and Assignment Operator. Decision Statements - if Statements, if-else Branching, switch Statements.

Case study

Library Management System

Scenario**:** A university library wants to modernize its book management system. They need a software solution to keep track of books, borrowers, and transactions efficiently.

Unit II Control Statements **9 Hrs**

Looping Statements: using for loop, using while Loops, Using do Loops. Jump Statements: using break and continue, Unlabelled Statements, Labelled Statements. Arrays: Declaration, Instantiation and Initialization of Java Array, Types of Arrays - Single Dimension array, Multi-dimension array -

Case study

Flight Reservation System

Scenario: An airline company wants to upgrade its flight reservation system to handle a large volume of bookings efficiently.

Unit III Strings **9 Hrs**

Strings: String, String Builder, and String Buffer, The String Class, Important Facts About Strings and Memory, Important Methods in the String Class, The string Buffer and String Builder Classes, Important Methods in the string Buffer and String Builder Classes, File Navigation and I/O.

Case Study

Banking application

Scenario**:** Develop an application for customer bank transaction management

Unit IV Streams **9 Hrs**

Types of Streams, The Byte-stream I/O hierarchy, Character Stream Hierarchy, Random Access File class, The java.io.Console Class, Serialization, Dates, Numbers, and Currency, Working with Dates, Numbers, and Currencies, Parsing, Tokenizing, and Formatting, Locating Data via Pattern Matching, Tokenizing

Case Study

Online Shopping Platform

Scenario: An e-commerce company wants to revamp its online shopping platform to provide a seamless shopping experience for customers.

Unit V Event handling **9 Hrs**

Basics of event handling – event handlers – adapter classes – actions – mouse events –AWT event hierarchy – Introduction to Swing – Model-View-Controller design pattern – buttons – layout management – Swing Components – exception handling – exception Hierarchy – throwing and catching exceptions.

Case Study

Intelligent Personal Assistant

Scenario: A java Application for Intelligent Personal Assistant to kept records of all the day to day activities.

Total Hours:	**45**

Laboratory Component:

S.No.	List of Experiments
1	Implement a Java program to model a simple real-world scenario using classes and objects, such as a banking system or a student information system.
2	Implement a linked list data structure in Java and perform operations like insertion, deletion, and traversal.
3	Use Weka or similar library to build and train a simple machine learning model for classification tasks, such as predicting iris flower species based on petal and sepal dimensions.
4	Create a web application using Java Servlets and JSP to manage a bookstore inventory, allowing users to search for books, add them to a shopping cart, and proceed to checkout.
5	Develop a web-based sentiment analysis tool using Java Servlets and JSP, where users can input text and get back the sentiment analysis result (positive, negative, or neutral).
6	Implement a multithreaded Java program to simulate a traffic intersection with multiple lanes, controlling the flow of vehicles using synchronized methods and locks.
7	Develop a concurrent web crawler in Java to crawl and index web pages from different domains simultaneously.
8	Use a Java library like Stanford NLP to perform advanced natural language processing tasks such as named entity recognition or sentiment analysis on text data. .
9	Integrate external APIs (e.g., Google Maps API) into a Java web application to provide functionalities such as location-based services or route planning.
10	Develop a Java application to stream and process real-time data from a source such as Twitter's streaming API, performing tasks like sentiment analysis or topic modeling on incoming tweets.
Project Work	
	Undertake a comprehensive AI project using Java, such as building a recommendation system for an e-commerce platform, a chatbot for customer support, or a predictive analytics tool for financial forecasting.
Text Books	

1.	Java: A Beginner's Guide" by Herbert Schildt (2018).
2.	Data Structures and Algorithms in Java" by Robert Lafore (2002).
3.	Machine Learning in Java" by Bostjan Kaluza (2018).
4.	Artificial Intelligence: A Modern Approach" by Stuart Russell and Peter Norvig (Latest Edition).
5.	Servlet and JSP: A Tutorial" by Budi Kurniawan (2022).
6.	Deeplearning4j: Cutting-edge Deep Learning for Java" by Adam Gibson and Josh Patterson (2017).
7.	Natural Language Processing in Java" by Richard M. Reese (2018).
Reference Books	
1.	Java: The Complete Reference" by Herbert Schildt (2024)
2.	Java Programming 24-Hour Trainer" by Yakov Fain (2021)
3.	Head First Java" by Kathy Sierra and Bert Bates (2021)
4.	Java Concurrency in Practice" by Brian Goetz, Tim Peierls, Joshua Bloch, Joseph Bowbeer, David Holmes, and Doug Lea (2020)
5.	Java Deep Learning Projects" by Md. Rezaul Karim and Md. Mahedi Kaysar (2023)
6.	Practical Java Machine Learning Projects with Google Cloud Platform and TensorFlow" by Harmeet Singh and Manpreet Singh Ghotra (2023)
7.	Java Deep Learning Essentials" by Yusuke Sugomori (2020)
8.	Java Deep Learning Cookbook" by Rick Bottini and Nick Gibson (2024)
9.	Machine Learning in Java" by Bostjan Kaluza (2021)
Web References	
1.	https://education.oracle.com/ko/java-developer-training-guide
2.	https://www.coursera.org/courses?query=machine%20learning&skills=Java%20Programming
3.	https://cloudxlab.com/course/188/pg-certificate-program-in-data-science-ai-by-cec-iit-roorkee
Online Resources	
1.	https://stackoverflow.com/
2.	https://github.com/

SEMESTER - IV

24UCSOE205	Real Time Embedded Systems	3/0/0
Nature of Course	Theory	
Prerequisites	NIL	

Course Objectives:

1	To consciousness among students towards different embedded applications around them.
2	To understand step by step process of designing embedded systems application using waterfall model.
3	To realize design of real time system using µCOS-II RTOS and inspect various designing issues of real time system with respect to specific applications.
4	To become competent in design of real time and non real time embedded system application using RTOS and Embedded Linux respectively.
5	To develop skills for implementing application specific embedded systems.

Course Outcomes (Theory)

Upon completion of the course, students shall have ability to

1	Explain different design aspects of embedded system	[U]
2	Foster step by step design process of embedded systems for specified application using waterfall model.	[A]
3	Explain the structure and working of real-time and non real-time Operating Systems used in embedded system design.	[U]
4	Implement formal methods of RTOS for the design of real-time systems.	[A]
5	Design embedded systems for different domestic and industrial application by applying waterfall model.	[A]

Course Contents

UNIT I Introduction to Embedded Systems **9 Hrs**

Introduction to Embedded Systems: definition, Applications and recent trends in embedded system, Block diagram, Architecture of embedded system, Characteristics, Classification, Design Metrics, Optimization of Design metrics, Key Design challenges, Design constraints.

UNIT II Embedded System Development Cycle **9 Hrs**

Embedded System Development Cycle: Requirement engineering, requirement Specification, Hardware-Software Partitioning, Hardware-Software co-design, Integration, Testing, Quality Assurance, Aesthetic Design, Maintenance, and Electronics Waste Management. Iterative model, Agile model, Kano's model of

customer satisfaction.

UNIT III Embedded System Software Design **9 Hrs**

Embedded System Software Design: Software Architectures of embedded System, Concept and necessity of RTOS, Types of RTOS, Non Real-time Operating Systems: Introduction to Embedded Linux, Embedded Linux architecture, Real Time OS: Introduction to Vxworks, Introduction to µCOS-II: Features, Architecture.

UNIT IV µCOS-II Real Time Operating System **9 Hrs**

sµCOS-II Real Time Operating System: File structure, Concept of Task, Assign static and dynamic priority to the tasks, Multitasking, Clock tick, Types of services: - System Services, Task management Services, Time management services, Shared resources/critical section of code, Concept of deadlock, Protection mechanisms, ITC/IPC Services.

UNIT V Applications of Embedded Systems **9 Hrs**

Applications of Embedded Systems: Digital Camera, Automatic Chocolate vending machine, smart Mobile Phone, Home automation, farm automation, industrial automation, Voice operated devices like Alexa and IoT devices

Total Hours:	**45**

Text Books

1.	Frank Vahid, Tony Givargis, "Embedded System Design: A Unified Hardware/Software Introduction",Wiley India,(2009) ISBN:- 978-81-265-0837-2.
2.	Andrew Sloss, "ARM System Developer's Guide: Designing and Optimizing System Software", Morgan Kaufmann, (2004), ISBN: 978-15-586-0874-0
3.	Jean J. Labrosse, "MicroC/OS-II: The Real Time Kernel", CRC Press; 2nd edition, (2002), ISBN: 978-15- 782-0103-7
4.	Christopher Hallinan, "Embedded Linux Primer: A Practical Real-World Approach", Prentice Hall, 2 edition, (2010).

Reference Books

1.	David E. Simon, "An Embedded Software Primer", Addison Wesley; 2nd edition, ISBN: 978-02-016-1569-2
2.	Raj Kamal, "Embedded Systems: Architecture, Programming and Design", Tata McGraw-Hill Education (India), (2011).
3.	Dr. K. V. Prasad, "Embedded Real Time Systems: Concepts, Design and Programming", Dreamtech Press, (2003) ISBN: 978-81-772-2461-0

Web References

1	NPTEL Course on RTOS https://nptel.ac.in/courses/106/105/106105172/
2	NPTEL Course on Embedded system design using ARM https://nptel.ac.in/courses/106/105/106105193/

3	Coursera course on Real Time Embedded System: https://www.coursera.org/specializations/real-time-embedded-systems
Online Resources	
1	https://www.micrium.com/rtos/kernels/
2	https://www.arduino.cc/
3	https://www.raspberrypi.org/

24UCSVS201	Compiler Construction Tools	2/0/0
Nature of Course:	Theory	
Prerequisites:	NIL	
Course Objectives:		
1	To teach concepts of language translation and phases of compiler design	
2	To describe the common forms of parsers	
3	To inculcate knowledge of parser by parsing LL parser and LR parser.	
4	To demonstrate intermediate code using technique of syntax directed translation	
5	To Illustrate the various optimization techniques for designing various optimizing compilers	
Course Outcomes		
1	Understand compiler construction and functionality.	[U]
2	Analyze different representations of intermediate code.	[A]
3	Optimize Syntactical Errors/Semantic errors during the compilation.	[A]
4	Design and construct new compiler for new languages.	[A]
5	Simulate Functionality of each stage of compilation process	[A]

UNIT I INTRODUCTION TO COMPILERS **6 Hrs**

Definition of compiler, interpreter and its differences, the phases of a compiler, role of lexical analyzer, regular expressions, finite automata, from regular expressions to finite automata, pass and phases of translation, bootstrapping, lexical analyzer generator, Parsing, role of parser, context free grammar, derivations, parse trees, ambiguity.

Case Study

Optimize the low-level intermediate to an equivalent level to C

UNIT II BOTTOM-UP PARSING **6 Hrs**

Definition of bottom-up parsing, handles, handle pruning, stack implementation of shift-reduce parsing, conflicts during shift-reduce parsing, LR grammars, LR parsers-simple LR, canonical LR(CLR) and Look Ahead LR (LALR) parsers, error recovery in parsing, parsing ambiguous grammars, YACC-automatic parser generator.

Case Study

Continuously improving the performance of generated code.

UNIT - III SYNTAX DIRECTED TRANSLATION: **6Hrs**

Syntax directed definition, construction of syntax trees, S-attributed and L-attributed definitions, translation schemes, emitting a translation. INTERMEDIATE CODE GENERATION: intermediate forms of source programs– abstract syntax tree, polish notation and three address code, types of three address statements andits implementation.

Case Study

Translates high-level source code into machine code using compiler.

UNIT - IV TYPE CHECKING: **6Hrs**

Definition of type checking, type expressions, type systems, static and dynamic checking of types, specification of a simple type checker, equivalence of type expressions, type conversions, overloading of functions and operators. RUN TIME ENVIRONMENTS: Source language issues, Storage organization, storage-allocation strategies.

Case Study

Converts source code into LLVM IR (Low-Level Virtual Machine).

UNIT - V CODE OPTIMIZATION: **6Hrs** Organization of code optimizer, basic blocks and flow graphs, optimization of basic blocks, the principal sources of optimization, the directed acyclic graph (DAG) representation of basic block, global data flow analysis. CODE GENERATION: Machine dependent code generation, object code forms, the target machine, a simple code generator, register allocation and assignment, peephole optimization. **Case Study** Dead Code Optimization or Dead Code Elimination.	
Total Hours:	**30**

Text Books:	
1.	Alfred V. Aho, Ravi Sethi, Jeffrey D. Ullman (2007), Compilers Principles, Techniques and Tools, 2nd edition, Pearson Education, New Delhi, India.
2.	"Modern Digital Electronics", R.P. Jain, 4 th Edition, (2009), Tata McGraw-Hill, ISBN 10: 0070669112 / ISBN 13: 9780070669116.
3.	M. Morris Mano, Michael D. Ciletti (2008), Digital Design, 4th edition, Pearson Education Inc, India.
Reference Books:	
1.	Alfred V. Aho, Jeffrey D. Ullman (2001), Principles of compiler design, Indian student edition, Pearson Education, New Delhi, India.
2.	Kenneth C. Louden (1997), Compiler Construction– Principles and Practice, 1st edition, PWS Publishing.
3.	K. L. P Mishra, N. Chandrashekaran (2003), Theory of computer science- Automata Languages and computation, 2nd edition, Prentice Hall of India, New Delhi, India.
4.	Andrew W. Appel (2004), Modern Compiler Implementation C, Cambridge University Press, UK.
Web References:	
1.	https://onlinecourses.nptel.ac.in/noc20_cs13/preview
2.	https://online.stanford.edu/courses/soe-ycscs1-compilers
Online Resources:	
1.	https://www.youtube.com/watch?v=iojJeCGC9nY

24UCSVE202	Foreign Language - German- II	0/0/4
Nature of Course	Practical	
Prerequisites	NIL	

Course Objectives

1	To develop all language skills (reading, listening, speaking and writing) at A2 level.
2	To develop communication skills in typical everyday situations.
3	To introduce basic vocabulary and grammatical structures to communicate in everyday situations.
4	To provide tools for further development of language skills, also in the academic context.
5	To comprehend and utilize key German phrases for everyday situations.

Course Outcomes
Upon completion of the course, students shall have ability to

1	Understand and identifies words and expressions connected with areas of most immediate personal relevance (family, everyday life, work).	[U]
2	Understand and identifies grammatical structures required at A2 level.	[U]
3	Utilize German vocabulary to discuss personal hobbies and interests in simple conversations.	[U]
4	Distinguish between different food and beverage options on a German menu.	[A]
5	Apply presents basic information connected with the geography and culture of a given country.	[A]

Laboratory Component

S.No.	Lab Exercise
1	Listen to a short German audio clip (e.g., a dialogue in a store) and answer comprehension questions about the content.
2	Use a language lab tool or software to record yourself reading a short German text. Compare your pronunciation with a native speaker's recording.
3	Complete interactive exercises focused on A2-level grammar topics such as past tense (Perfekt), modal verbs, and accusative and dative cases.
4	Use flashcard software to learn and practice vocabulary related to common themes like travel, food, and daily activities. Include exercises that require you to use new words in sentences.
5	Pair up with a partner and practice role-playing common situations (e.g., booking a hotel room, shopping for groceries). Record and review the dialogues for feedback.
6	Write a short essay or a series of sentences on a given topic (e.g., describe your family, write about your last holiday). Use software that can provide instant grammar and spelling feedback.
7	Read a short German text (e.g., a news article or a story). Answer questions about the main ideas, details, and vocabulary used in the text.
8	Use a language learning platform that offers interactive storytelling exercises. Choose your own adventure-style stories in German and make decisions to advance the plot.
9	Participate in a virtual language exchange session with native German speakers or other learners. Practice speaking on various topics and receive real-time feedback.
10	Watch a short German film or video clip. Discuss the cultural aspects you observed, such as traditions, holidays, or typical daily routines, and compare them to your own culture.

Text Books

1.	Deutsch im Blick - Strong foundation in grammar and vocabulary for reading comprehension,(2013).
2.	Complete German - Structured approach with interactive exercises.
3.	Menschen A1.1 - Communicative approach for real-life situations,(2015).
Reference Books	
1.	Langenscheidt Dictionary - Comprehensive German-English dictionary.
2.	New German Grammar - Detailed explanations of grammar rules.
Web References	
1.	Goethe-Institut Online Courses - Interactive exercises and grammar explanations.
2.	DW Learn German - Video lessons, podcasts, and vocabulary trainers.
Online Resources	
1.	Memrise - Flashcards for memorizing vocabulary.
2.	Duolingo - Gamified app for basic phrases and grammar.

24UCSVS202	Community Engagement Mini Project I	0/0/4
Nature of Course:	Programming	
Prerequisites:	NIL	
Course Objectives:		
1	To understand and implement fundamental and advance concepts of software engineering	
2	To provide a strong foundation in project development and practices near to real time problem solving.	
3	To give systematic design approaches covering conceptual, logical and physical design of mini project.	
4	To learn and understand methods, architectures and real time applications as problem solving.	
5	To learn a powerful, flexible and scalable general purpose application to handle all types data.	
Course Outcomes **Upon completion of the course, students shall have ability to:**		
1	Understand & Implement fundamental concepts of software engineering to solve real world problem.	[U]
2	Understand project development life cycle and various best practices of project development.	[U]
3	Implement mini project using systematic design approaches.	[AP]
4	Apply different methods, architectures and real time application of problem solving.	[AP]
5	Compare different type of data and use of appropriate Architecture, method to develop solution in real time.	[A]

Course Contents

General Guidelines for Mini-Project

1. The mini project should be undertaken preferably by a group of 2-3 students under the guidance of domain expert faculty who will jointly work together and implement the project.
2. Topics should be based on the technology that students have studied in choice based subjects.
3. It is appreciated if the mini-project is based on real world problems and day to day life.

Use of open source software is to be appreciated.

1. The group has to select the project topic with the approval of the guide and submit the name of the project with synopsis of the proposed work.
2. At the end of the semester each group need to submit a report of minimum 25 pages.

Guidelines for Assessment

Continuous assessment of laboratory work is done based on overall performance in lab assignments and mini-project. Each lab assignment assessment will assign grade/marks based on parameters with appropriate weightage.

Term Work will be based on assignments carried out by a student and mini-project demonstration and related skills learned.

The formats for synopsis and report are as given below:

1. Title page 1
2. Certificate Page 1
3. Acknowledgement Half
4. Content (Index) 1 or one & half
5. Introduction 1 or 2
6. Problem statement and Objectives 1
7. Literature Review 5 to 10 pages depending on topic
8. Work Methodology as per nature of topic/work 10 to 20 pages
9. Summary of Results and discussion 2 to 4 pages
10. Conclusion and Future Scope 1 or half page
11. Letter of interaction with outside industry /institute/ individuals, paper details in journal or Participation certificate in conference, seminar, paper, Email correspondence, 1 to 4 pages
12. References 1 or 2 pages
13. Appendix if any 1 to 4 pages.

Procedure of Evaluation

Normally, evaluation of mini-projects is done through presentations by a group of students in front of two or more faculty and assessment of individual students is done by faculty and average of marks are worked out. Proper Assessment Rubrics will be developed and disseminated by the faculty. General procedure is given below.

1. As per departmental academic policy, mini-projects may be evaluated similar to laboratory course work.
2. The faculty Coordinator collects the soft copies of Mini-projects in the department and the grouping of the Mini-projects is done depending upon the topics of the Mini projects. Panels of 3 to 4 faculty guiding concerned Mini-projects, are formed for the evaluation
3. Mini-project Guide (internal Faculty) will evaluate project reports submitted by his group of students, in the form of soft copy in the "suggested" format and recalling the observations of performance of the students in a group, faculty will give marks out of 25.
4. As per the midterm evaluation schedule, a concerned panel of the faculty does the evaluation of the Mini-projects and average marks out of 25 are given to the students. Suggestions by the panels during the presentation works play a very important role for the motivation and guidance for further work on the Mini –projects.
5. After mid semester evaluation for 25 marks, student groups continue their work under the guidance of concerned guides. After completion of the Mini-projects by the students, the students under the guidance of concerned guides prepare a small report based on the work and the faculty guide does the evaluation for 25 marks.
6. The final presentation (Internal Evaluation) of the Mini-projects for 50 marks will be conducted as a part of external ORAL Examination.

Common Instruction

1. Department has to identify skill sets requirements in consultation with Industry.
2. The mini-project will be on a particular skill set only, it is encouraged to conduct it with the involvement of Industry Expert to acquire such skill set, and internal faculty will act as a facilitator for the students.
3. Proper Assessment Rubrics will be developed, explained to the students, disseminated to the students well in advance.
4. At the end of this, students must be able to exhibit the acquired skills through its proper use in the development of selected applications.

24UCSMD202	Business Analytics using SAS	2/0/0

Nature of Course	Theory
Prerequisites	NIL

Course Objectives:

1	Gain a fundamental understanding of Business Analytics and proficiency in navigating SAS software.
2	Develop skills in data cleaning, preparation, and manipulation using SAS functions.
3	Apply descriptive and inferential statistical methods to analyze and interpret data in SAS.
4	Build and validate basic predictive models using SAS for forecasting and decision-making.
5	Explore advanced analytics techniques, including clustering and data visualization, to derive actionable insights from data.

Course Outcomes
Upon completion of the course, students shall have ability to

1	Demonstrate comprehension of fundamental concepts of Business Analytics and SAS.	[U]
2	Apply data manipulation techniques effectively to clean, transform, and manage datasets in SAS.	[AP]
3	Analyze data using descriptive and inferential statistical methods to draw conclusions and make informed decisions.	[A]
4	Create and evaluate predictive models using appropriate statistical techniques in SAS.	[A]
5	Synthesize and present complex data insights using advanced analytics techniques, such as clustering and visualization, to inform strategic decision-making.	[A]

Unit I Introduction to Business Analytics and SAS **6 Hrs**
Overview of Business Analytics, Importance and Applications of Business Analytics, Introduction to SAS, Installing and Navigating SAS Software, Basic SAS Syntax, Importing and Exporting Data.
Case Study
Analyze a small dataset using basic SAS commands to generate summary statistics and present your findings briefly.

Unit II Data Manipulation and Management **6 Hrs**
Data Cleaning and Preparation, Handling Missing Values, Using SAS Functions for Data Manipulation, Sorting and Filtering Data, Merging and Appending Datasets.
Case Study
Clean and prepare a provided dataset, handle missing values, and perform basic data transformations. Document the steps taken.

Unit III Descriptive and Inferential Statistics **6 Hrs**
Descriptive Statistics: Mean, Median, Mode, Variance, Inferential Statistics: Hypothesis Testing, t-tests, Chi-Square Tests, Using SAS Procedures for Statistical Analysis.
Case Study
Conduct a simple hypothesis test using SAS on a given dataset. Present your process and results in a concise report.

Unit IV Predictive Modeling and Forecasting **6 Hrs**
Introduction to Predictive Modeling, Linear Regression, Logistic Regression, Time Series Analysis, Using SAS Procedures for Predictive Modeling.
Case Study
Develop a basic predictive model using linear regression on a dataset. Validate the model and summarize its accuracy briefly.

Unit V Advanced Topics in Business Analytics **6 Hrs**
Introduction to Machine Learning, Clustering Techniques, Data Visualization Techniques, Creating Graphs in SAS, Case Studies.

Case Study Perform a clustering analysis on a dataset to identify patterns. Visualize the results and provide a short summary of your findings.	
Total Hours	**30**
Text Books	
1.	Nathaniel Lin, Applied Business Analytics: Integrating Business Process, Big Data, and Advanced Analytics", Pearson Education, (2015).
2.	Ron Cody, "Learning SAS by Example: A Programmer's Guide", SAS Institute, (2016).
3.	James R. Evans, "Business Analytics: Methods, Models, and Decisions", Pearson Education, (2016).
Reference Books	
1.	Gareth James, "An Introduction to Statistical Learning: with Applications in R", Springer, (2013).
2.	Jeffrey D. Camm et al., "Business Analytics: Data Analysis & Decision Making", Cengage Learning, (2020).
Web References	
1.	SAS Official Documentation: https://documentation.sas.com/
2.	Coursera - Business Analytics Specialization: https://www.coursera.org/specializations/business-analytics
Online Resources	
1.	Towards Data Science - Articles on Data Analysis and Business Analytics: https://towardsdatascience.com/tagged/business-analytics
2.	SAS Communities - Online Forums for SAS Users: https://communities.sas.com/

24UCYPC205	Database Management System	3/0/2
Nature of Course	Theory & Programming	
Prerequisites	NIL	

Course Objectives

1	To understand to design entity-relationship diagrams to represent simple database application scenarios
2	To understand how to convert entity-relationship diagrams to relational- and graph-oriented implementations
3	To understand the fundamental concepts for data manipulation with queries.
4	To understand that no single data architecture can be used to meet all data management requirements
5	To understand the transaction management in the database area.

Course Outcomes (Theory) - Upon completion of the course, students shall have ability to

1	Apply design entity-relationship diagrams to represent simple database application scenarios	[AP]
2	Apply how to convert entity-relationship diagrams to relational- and graph-oriented implementations	[AP]
3	Apply the fundamental concepts for data manipulation with queries.	[AP]
4	Apply storing data methods to meet all data management requirements	[AP]
5	Understand with the transaction management in the database area.	[U]

Course Contents

Unit 1 Database Design **9Hrs**

File system vs a DBMS, Queries in a DBMS, Transaction Management, Database design and ER diagram, Entities, Attributes and Entity Sets, Relationships and Relationship Sets, Features of the ER Model, Conceptual Design with the ER Model, Conceptual Design for Large Enterprises, The Unified Modeling Language.

Case Study

The Internet Store

Unit II The Relational Model **9 Hrs**

Introduction to the Relational Model, Integrity Constraints over Relations, Enforcing Integrity Constraints, Querying Relational Data, Logical Database Design: ER to Relational, Introduction to Views, Destroying/Altering Tables and Views**,** Normal forms, Multivalued Dependencies, Inclusion Dependencies. Schema Refinement in Database Design.

Case Study

The Internet Store

Unit III Relational Algebra, Calculus & SQL Queries **9 Hrs**

Relational Algebra, Relational Calculus, SQL Query, Aggregate Operators, Null values, Complex Integrity Constraints in SQL, Triggers and active database, Designing active databases. NoSQL, MySQL.

Case Study

The Internet Store

Unit IV Storing Data: Disks and Files **9 Hrs** The Memory Hierarchy, Redundant Arrays of Independent Disks, Disk Space Management, Buffer Manager, Files of Records, Page Formats, Record Formats, Indexed Sequential Access Method (ISAM), B+ Trees, Hash-Based Indexing, MongoDB, IPFS (Inter Planetary File System). **Case Study** The Internet Store **Unit V Transaction Management –** **9 Hrs** The ACID Properties, Concurrent Execution of Transactions, Lock-Based Concurrency control, Transaction support in SQL, Introduction to Crash Recovery, 2PL, Serializability and Recoverability, Lock Management, Dealing with Deadlocks, Specialized Locking Techniques, ConCurrency Control without Locking. **Case Study** The Internet Store	
Total Hours	**45**

Laboratory Assignments	
1	Create an E R diagram and relational schema to hold information about the situation in many institutions affiliated to some University, many teachers of different disciplines are teaching to many students enrolled in many courses offered by the university to the students through the institutions. Use concept of keys, aggregation, generalization, cardinality etc. in a proper way. Say the schema of respective entities is: • Teacher(T#, Tname, Tqual, Tsubject, Tcourse, Prog) • Student(Roll#., Sname, Sage, Saddress, Scourse.Prog , Smarks) • Teaches(T#, Roll# , Scourse, Prog ,University)
2	The NBA (NDHU Basketball Association) is in dire need of a database in order to keep track of the activities in their league. The entities in the database are • People (with attributes id, name and age), • Teams (with attributes team name and manager), and • Courts (with attributes court id, address and capacity). Furthermore, people are specialized into Referees and Players. • Referees have the extra attribute level and • Players have the extra attribute height. • Players play in teams, and teams and referees participate in a game that takes place in a court on a certain date. This league is quite violent and very often players are fouled out by referees. This causes bad relationships between teams and referees and some teams disqualify some referees from refereeing their games. Some way of recording each game is also required. This will need to store the home and away teams scores. No player plays for more than one team. Only one game is played on one court on any one day. a. Produce a set of normalized entities to 1NF. Add attributes as you think they are required. b. Create an ER diagram that models the relationships in the system.
3	The following tables form part of a database held in a relational DBMS: • Hotel (Hotel_No , Name, Address) • Room (Room_No, Hotel_No, Type, Price) • Booking (Hotel_No, Guest_No, Date_From, Date_To, Room_No) • Guest (Guest_No, Name, Address) Where Hotel contains hotel details and Hotel_No is the primary key. Room contains room details for each hotel and (Hotel_No, Room_No) forms the primary key. Booking contains details of the bookings and the primary key comprises (Hotel_No, Guest_No, Date From) and Guest contains guest details and Guest_No is the primary key. Write the SQL statements for the following: a. List full details of all hotels in Shirdi. b. List the names and addresses of all guests in Kolkata, alphabetically ordered by name. c. List all double or family rooms with a price below Rs.1000 per day, in ascending order of price. d. List the bookings for which no date to has been specified. e. What is the total daily revenue from all the double rooms? f. How many different guests have made bookings for August, 2023? g. List the price and type of all rooms at the hotel Land Mark. h. What is the total income from bookings for the hotel Sai-Maya Today?
4	Consider the following requirements for a construction company that uses database system to keep track of its projects, workers and material requirements of various projects. The projects for the company are of two kinds: (i) Turn key projects and (ii) Others. All the projects have a life cycle (Please note that the turn key projects have a maintenance phase in addition). And workers are allotted as per the phase of the project. Each project has its own plan of completion that is drawn at the start of the project. The worker and material requirement of project is calculated at the start of the project.

	The store manages the materials. One of the major constraints for the store is that it requires 15 days to acquire a product. Thus, the inventory should fulfill the requirements of the next 15 days. The store also keeps track of the issue of materials and return of defective materials to various projects. The company keeps the information of all the clients including the information about the projects that are being maintained by the company (turn key projects). Draw an ER diagram for the above company. Specify key attributes and constraints of each entity type and of each relationship type.
5	Assume that a Consumer item lease Company which leases various household items to its clients for their use for a specific period of time, maintains the following tables: • Clients (clientID, name, address, contact Phone) • Itemlist (itemID, itemName, itemCost, purchase Date) • Leaselist (clientID, transactionNO, itemID, startDate, returnDate, amountTObeCharged) Write and run the following SQL queries on the tables: a. Find all the client names that have not got any item leased during the last month and no leased item is pending with them. b. Find the list of all the items that were leased or returned last month. c. Find the names of all those clients who have given the business to the company in the decreasing order of total amount paid by a client. d. List the client's details and the items leased to them at present. e. Find the client who has been leased at least two items.
Text Books	
1	Ramakrishnan, Raghu, and Johannes Gehrke. Database Management Systems. 3rd ed. McGraw-Hill, 2002. ISBN: 9780072465631.
2	Garcia-Molina, Ullman, Widom. Database Systems: The Complete Book. Prentice Hall, 2002 (or 2008, 2nd ed).
Reference Books	
1	Hellerstein, Joseph, and Michael Stonebraker. Readings in Database Systems (The Red Book). 4th ed. MIT Press, 2005. ISBN: 9780262693141.
2	Lemahieu, W., Broucke, S. van den and Baesens, B. (2018) Principles of database management. Cambridge University Press.
3	Database System Concepts (6th Edition) by Silberschatz, Korth, and Sudarshan.
4	Joseph M. Hellerstein, Michael Stonebraker, and James Hamilton Architecture of a Database Foundations and Trends in Databases Vol. 1, No. 2 (2007) 141–259
Web References	
1	MIT Open Course Ware – https://ocw.mit.edu/courses/6-830-database-systems-fall-2010/
2	Cambridge University- https://www.cl.cam.ac.uk/teaching/2223/Databases/
3	Imperial College London- https://imperial.alma.exlibrisgroup.com/leganto/readinglist /lists/43883457220001591?institute=44IMP_INST&auth=SAML
4	ETH Zurich – https://systems.ethz.ch/education/courses/2022-autumn-semester/data-management-systems.html
Online Resources	
1	Swayam – https://onlinecourses.nptel.ac.in/noc19_cs46/preview
2	Swayam – https://onlinecourses.nptel.ac.in/noc24_cs55/preview
3	University of Colorado- https://www.coursera.org/learn/database-management
4	Meta- https://www.coursera.org/learn/database-structures-and-management-with-mysql

24UCSPC206	Theory of Computer Science	3/0/0
Nature of Course:	Theory	
Prerequisites:	Discrete Structure	
Course Objectives:		
1	To study Finite State Machine, Finite Automata and its language	
2	To learn Regular Expressions and Regular Languages	
3	To understand Context Free Grammars and Context Free Languages	
4	To study Pushdown Automata and its language	
5	To learn and understand Turing Machine and its complexity	
Course Outcomes		
Upon completion of the course, students shall have ability to		
1	Construct Finite Automata and its variants for regular languages	[U]
2	Build regular expressions for a regular language and to prove theorems and properties of regular languages.	[AP]
3	Write context free grammar for context free languages and to prove properties of CFL.	[AP]
4	Construct Pushdown Automata for context free language	[AP]
5	Construct Turing Machines for unrestricted kind of languages considering complexity	[AP]

Unit I Finite Automata **9 Hrs**

Finite automata, Basic notation and convention DFA, Examples of DFA, Computation by DFA and regular operation, Nondeterminism, NFA, Equivalence of NFA and DFA, Regular expression, RE to NFA, GNFA to RE, Kleene's Theorem, Pumping Lemma, Myhill-Nerode Theorem.

Case Study

Traffic Lights, Video Games, Text Parsing designing using FSM.

Unit II Regular Expression **9 Hrs**

Operators of RE, Building RE, Precedence of operators, Algebraic laws for RE, Equivalence of two RE's, Regular expressions vs. FAs, DFA to RE using Arden's theorem, Loop elimination method, Minimization of DFA, Equivalence of DFA, Closure and Decision properties.

Case Study

Street address designing and roman numerals using regular expression.

Unit III Context Free Grammar and Languages **9 Hrs**

Context Free Grammar, sentential forms, Derivations, Parse trees, Context Free Language. Ambiguous Grammar, Inherent ambiguity, writing a grammar for language. Simplification of CFG, eliminating Є-productions, unit productions, useless production, useless symbols, CYK Algorithm, Right and Left linear grammar. Normal Forms- Chomsky normal form, Greibach normal form, Closure properties of CFL, Decision properties of CFL's,

Case Study

Human behavior recognition using a context-free grammar.

Unit IV Push Down Automata and Post Machine **9 Hrs**

Formal Definition of the PDA, Equivalence of Acceptance by Final State & Empty stack, Non-Deterministic PDA, PDA and Context Free Language, Introduction to Deterministic and Nondeterministic Pushdown Automata. Equivalence of PDA's and CFG's, Post Machine, Parsing Techniques.

Case Study

Syntactic Analysis (Parsing): use of PDA to turn a steam of tokens into an Abstract Syntax Tree (AST).

Unit V Turing Machine and Complexity **9 Hrs**

Turing Machine Model, Formal Definition of TM, Instantaneous description for TM, Transition diagrams for TM, The Language of Turing Machine, Design of TM, Description of TM,
Hierarchy theorems, complexity class P, NP class, NP-completeness, Space complexity: PSPACE, NPSPACE, PSPACE-completeness, L, NL, NL-Completeness

Case Study

Approximating Turing Machines with Transformers.

Total Hours	**45**

Text Books	
1.	John E. Hopcroft, Rajeev Motwani, Jeffrey D.Ullman, "Introduction to Automata Theory Languages and Computation", Addison-Wesley, ISBN 0-201-44124-1(2008).
2.	John Martin, "Introduction to Languages of The Theory of Computation", 2nd Edition, McG Hill Education, ISBN-13: 978-1-25-900558-9(2017).
3.	H.L. Lewis, Christos H. Papadimitriou, "Elements of the Theory of Computation", Prentice Hall, ISBN-10: 0132624788; ISBN-13: 978-0132624787(2015).
Reference Books	
1.	Michael Sipser, "Introduction to the Theory of Computation", Cengage Learning, ISBN-13:97811331878137(2012).
Web References	
1.	https://cglab.ca/~michiel/TheoryOfComputation/TheoryOfComputation.pdf
2.	http://staff.ustc.edu.cn/~huangwc/book/Sipser_Introduction.to.the.Theory.of.Computation.3
3.	https://cs-people.bu.edu/mbun/courses/332_F21/
Online Resources	
1.	https://nptel.ac.in/courses/106/104/106104148/
2.	https://nptel.ac.in/courses/106/104/106104028/